

# C v3 Vertical Slice v1 - Stage 2 Semantic Repair Human Review

Repair validation: PASS | Human semantic audit: PENDING | Stage 3: NOT STARTED

## 20-point regression overview

- LI18 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- ST10 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- LI17 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- BL17 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- BL23 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- BL25 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- ST9 | RESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- TE20 | UNRESOLVED -> UNRESOLVED | pre-repair rules=[] | post-repair findings=0
- HT7 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- LI4 | RESOLVED -> UNRESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'parent\_only\_composite\_resolution', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- ST1 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- GB14 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- GB23 | RESOLVED -> RESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- LU6 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- LI7 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- GB26 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- ST2 | UNRESOLVED -> UNRESOLVED | pre-repair rules=['operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- BL40 | RESOLVED -> RESOLVED | pre-repair rules=['operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- ST4 | RESOLVED -> UNRESOLVED | pre-repair rules=['condition\_linkage\_loss', 'direction\_preservation\_failure', 'operand\_trace\_incomplete', 'parent\_only\_composite\_resolution', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0
- TE6 | RESOLVED -> RESOLVED | pre-repair rules=['direction\_preservation\_failure', 'operand\_trace\_incomplete', 'relation\_nonoperand\_semantics\_loss', 'relation\_semantic\_hash\_mismatch'] | post-repair findings=0

# LI18: RESOLVED -> UNRESOLVED

Defect family: composite\_subfeature\_collapse, modifier\_dependency\_loss

## WHO source

location | S:LI18:location | page 51 | On the anterior aspect of the neck, at the same level as the superior border of the thyroid cartilage, between the anterior and posterior borders of the sternocleidomas-toid muscle.

## Semantic decomposition

LM:S:LI18:location:115-180:0f3e9745 | source=anterior and posterior borders of the sternocleidomas-toid muscle | class=muscle | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI18:location:128-180:9046d75e | source=posterior borders of the sternocleidomas-toid muscle | class=muscle | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI18:location:57-101 | source=the superior border of the thyroid cartilage | class=cartilage | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI18:location:61-76 | source=superior border | class=boundary.border | parent=LM:S:LI18:location:84-101 | child=superior border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no approved executable construction rule for subfeature 'superior border'

LM:S:LI18:location:7-22 | source=anterior aspect | class=surface.aspect | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"71f7a5ab627b3c93a9aceb574aadbdb27ca3577688316b7de105a5effa7840dc7","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:anterior\_neck","subfeature\_type":null} | reason=None

LM:S:LI18:location:84-101 | source=thyroid cartilage | class=cartilage | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA55099","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

## Composite binding view

PARENT LM:S:LI18:location:84-101 thyroid cartilage status=RESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA55099","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:LI18:location:61-76 superior border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

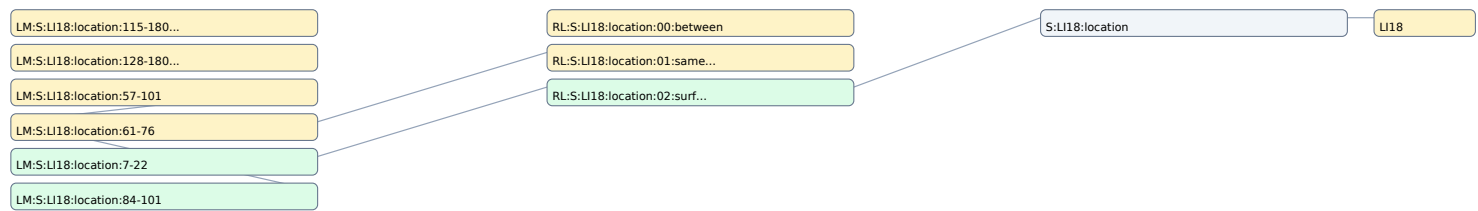
## Relation execution

RL:S:LI18:location:00:between | op=between\_constraint | semantic=['LM:S:LI18:location:115-180:0f3e9745', 'LM:S:LI18:location:128-180:9046d75e'] | executable=[] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:LI18:location:01:same-level | op=same\_level\_plane | semantic=['LM:S:LI18:location:57-101'] | executable=['LM:S:LI18:location:61-76'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:LI18:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:LI18:location:7-22'] | executable=['LM:S:LI18:location:7-22'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

- LM:S:LI18:location:57-101 -> LM:S:LI18:location:61-76 [semantic\_to\_executable]
- LM:S:LI18:location:61-76 -> RL:S:LI18:location:01:same-level [executable\_to\_relation]
- LM:S:LI18:location:7-22 -> RL:S:LI18:location:02:surface-landmark [executable\_to\_relation]
- RL:S:LI18:location:02:surface-landmark -> S:LI18:location [relation\_to\_statement]
- LM:S:LI18:location:84-101 -> LM:S:LI18:location:61-76 [parent\_to\_child\_subfeature]
- S:LI18:location -> LI18 [statement\_to\_point]

## BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:LI18:location:115-180:0f3e9745","LM:S:LI18:location:128-180:9046d75e"],"branch\_id":null,"op":"between\_constraint","source\_statement\_id":"S:LI18:location"},"executor":"between\_constraint","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI18:location:57-101"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:LI18:location"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI18:location:7-22"],"branch\_id":null,"op":"surface\_membership","source\_statement\_id":"S:LI18:location"},"executor":"surface\_membership","status":"RESOLVED"}]}

S:LI18:location"},"executor":"surface\_membership","status":"RESOLVED"}], "status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"executable\_argument\_node\_ids":[],"executor":"between\_constraint","operand\_binding\_trace":[{"binding":"missing\_executable\_subfeature","executable\_operand\_ids":[],"semantic\_operand\_id":"LM:S:LI18:location:115-180:0f3e9745"}, {"binding":"missing\_executable\_subfeature","executable\_operand\_ids":[],"semantic\_operand\_id":"LM:S:LI18:location:128-180:9046d75e"}], "reason":"semantic composite has multiple child subfeatures and no unique executable operand","semantic\_argument\_node\_ids":["LM:S:LI18:location:115-180:0f3e9745","LM:S:LI18:location:128-180:9046d75e"],"semantic\_fields":{"branch\_id":null,"op":"between\_constraint","relation\_type":"between","semantic\_argument\_node\_ids":["LM:S:LI18:location:115-180:0f3e9745","LM:S:LI18:location:128-180:9046d75e"],"source\_statement\_id":"S:LI18:location"},"semantic\_fields\_hash":"53fac0924ad39668b91eeae05f11db16956d7d7b62ecec90aae94ea57123330d","status":"UNRESOLVED"}], {"executable\_argument\_node\_ids":["LM:S:LI18:location:61-76"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"child\_subfeature","executable\_operand\_id":"LM:S:LI18:location:61-76","semantic\_operand\_id":"LM:S:LI18:location:57-101"}], "reason":"executable operand not uniquely executable","semantic\_argument\_node\_ids":["LM:S:LI18:location:57-101"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:LI18:location:57-101"],"source\_statement\_id":"S:LI18:location"},"semantic\_fields\_hash":"cc35e86561e20ead9da852c408f0987a243ed966cd52a30fb4f79a2b8b7314b3","status":"UNRESOLVED"}], {"constraint":{"argument\_node\_ids":["LM:S:LI18:location:7-22"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:LI18:location:7-22"],"op":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI18:location:7-22","semantic\_operand\_id":"LM:S:LI18:location:7-22"}], "relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:LI18:location:7-22"],"source\_statement\_id":"S:LI18:location"},"executable\_argument\_node\_ids":["LM:S:LI18:location:7-22"],"executor":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI18:location:7-22","semantic\_operand\_id":"LM:S:LI18:location:7-22"}], "semantic\_argument\_node\_ids":["LM:S:LI18:location:7-22"],"semantic\_fields":{"branch\_id":null,"op":"surface\_membership","relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:LI18:location:7-22"],"source\_statement\_id":"S:LI18:location"},"semantic\_fields\_hash":"e73d4814e6128b34f4e17cad65b5956d7b7c58c3c3d0586f83cfb3a558c46b9f","status":"RESOLVED"}], "status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature
- [ ] distinct subfeatures
- [ ] relation operator
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# ST10: RESOLVED -> UNRESOLVED

Defect family: composite\_subfeature\_collapse

## WHO source

location | S:ST10:location | page 59 | In the anterior region of the neck, at the same level as the cricoid cartilage, just anterior to the border of the sternocleido-mastoid muscle.

## Semantic decomposition

LM:S:ST10:location:100-107 | source= border | class=boundary.border | parent=LM:S:ST10:location:111-142 | child= border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no approved executable construction rule for subfeature 'border'

LM:S:ST10:location:111-142 | source=the sternocleido-mastoid muscle | class=muscle | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13407","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST10:location:57-78 | source=the cricoid cartilage | class=cartilage | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA9615","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST10:location:7-34 | source=anterior region of the neck | class=region.body\_region | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"71f7a5ab627b3c93a9aceb574aadb27ca3577688316b7de105a5effa7840dc7","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:anterior\_neck","subfeature\_type":null} | reason=None

LM:S:ST10:location:96-142 | source= the border of the sternocleido-mastoid muscle | class=muscle | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

## Composite binding view

PARENT LM:S:ST10:location:111-142 the sternocleido-mastoid muscle status=RESOLVED  
identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13407","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:ST10:location:100-107 border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

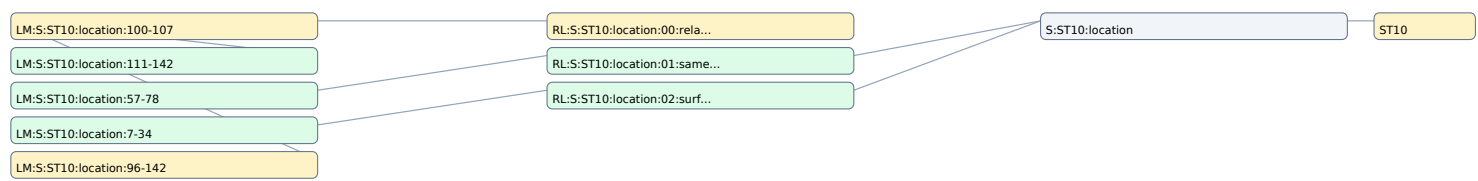
## Relation execution

RL:S:ST10:location:00:relative-to | op=directional\_relation | semantic=['LM:S:ST10:location:96-142'] | executable=['LM:S:ST10:location:100-107'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST10:location:01:same-level | op=same\_level\_plane | semantic=['LM:S:ST10:location:57-78'] | executable=['LM:S:ST10:location:57-78'] | dir=None | status=RESOLVED | reason=None

RL:S:ST10:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:ST10:location:7-34'] | executable=['LM:S:ST10:location:7-34'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

- LM:S:ST10:location:96-142 -> LM:S:ST10:location:100-107 [semantic\_to\_executable]
- LM:S:ST10:location:100-107 -> RL:S:ST10:location:00:relative-to [executable\_to\_relation]
- LM:S:ST10:location:57-78 -> RL:S:ST10:location:01:same-level [executable\_to\_relation]
- RL:S:ST10:location:01:same-level -> S:ST10:location [relation\_to\_statement]
- LM:S:ST10:location:7-34 -> RL:S:ST10:location:02:surface-landmark [executable\_to\_relation]
- RL:S:ST10:location:02:surface-landmark -> S:ST10:location [relation\_to\_statement]
- LM:S:ST10:location:111-142 -> LM:S:ST10:location:100-107 [parent\_to\_child\_subfeature]
- S:ST10:location -> ST10 [statement\_to\_point]

## BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:ST10:location:96-142"],"branch\_id":null,"direction":"anterior","op":"directional\_relation","source\_statement\_id":"S:ST10:location"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST10:location:57-78"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:ST10:location"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST10:location:7-34"],"branch\_id":null,"op":"surface\_membership","source\_statement\_id":"S:ST10:location"},"executor":"surface\_membership","status":"RESOLVED"}],"status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"executable\_argument\_node\_ids":["LM:S:ST10:location:100-107"],"executor":"directional\_relation","operand\_binding\_trace":[{"binding":"child\_subfeature","executable\_operand\_id":"LM:S:ST10:location:100-107","semantic\_operand\_id":"LM:S:ST10:location:96-142"}],"reason":"executable operand not uniquely executable","semantic\_argument\_node\_ids":["LM:S:ST10:location:96-142"],"semantic\_fields":{"branch\_id":null,"direction":"anterior","op":"directional\_relation","relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:ST10:location:96-142"],"source\_statement\_id":"S:ST10:location"},"semantic\_fields\_hash":"87063103c1542329c9be236d77c686f7ea4bc641f330022b473e4b197d4ffb94","status":"UNRESOLVED"},{"constraint":{"argument\_node\_ids":["LM:S:ST10:location:57-78"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:ST10:location:57-78"],"op":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST10:location:57-78","semantic\_operand\_id":"LM:S:ST10:location:57-78"}],"relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:ST10:location:57-78"],"source\_statement\_id":"S:ST10:location"},"executable\_argument\_node\_ids":["LM:S:ST10:location:57-78"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST10:location:57-78","semantic\_operand\_id":"LM:S:ST10:location:57-78"}],"semantic\_argument\_node\_ids":["LM:S:ST10:location:57-78"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:ST10:location:57-78"],"source\_statement\_id":"S:ST10:location"},"semantic\_fields\_hash":"1cfdcdcbac4b4281402e748a91faa05e0caf9e3c920bfef25618c01d1dc82b63","status":"RESOLVED"},{"constraint":{"argument\_node\_ids":["LM:S:ST10:location:7-34"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:ST10:location:7-34"],"op":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST10:location:7-34","semantic\_operand\_id":"LM:S:ST10:location:7-34"}],"relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:ST10:location:7-34"],"source\_statement\_id":"S:ST10:location"},"executable\_argument\_node\_ids":["LM:S:ST10:location:7-34"],"executor":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST10:location:7-34","semantic\_operand\_id":"LM:S:ST10:location:7-34"}],"semantic\_argument\_node\_ids":["LM:S:ST10:location:7-34"],"semantic\_fields":{"branch\_id":null,"op":"surface\_membership","relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:ST10:location:7-34"],"source\_statement\_id":"S:ST10:location"},"semantic\_fields\_hash":"e53fc06ca2764ec17cc7d12f4fbab5ededeb2155131962b8d858f38d0f8f8396","status":"RESOLVED"}],"status":"UNRESOLVED"}]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature
- [ ] distinct subfeatures
- [ ] relation operator
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# LI17: RESOLVED -> UNRESOLVED

Defect family: composite\_subfeature\_collapse

## WHO source

location | S:LI17:location | page 51 | On the anterior aspect of the neck, at the same level as the cricoid cartilage, just posterior to the border of the sternocleido-mastoid muscle.

note | S:LI17:note:1 | page 51 | Note: Directly inferior to LI18, at the same level as St10.

## Semantic decomposition

LM:S:LI17:location:101-108 | source= border | class=boundary.border | parent=LM:S:LI17:location:112-143 | child= border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no approved executable construction rule for subfeature 'border'

LM:S:LI17:location:112-143 | source=the sternocleido-mastoid muscle | class=muscle | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:LI17:location:57-78 | source=the cricoid cartilage | class=cartilage | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:LI17:location:7-22 | source=anterior aspect | class=surface.aspect | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:LI17:location:97-143 | source= the border of the sternocleido-mastoid muscle | class=muscle | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI17:note:1:26-31 | source= LI18 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:LI17:note:1:54-58 | source=St10 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

## Composite binding view

PARENT LM:S:LI17:location:112-143 the sternocleido-mastoid muscle status=RESOLVED  
identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:LI17:location:101-108 border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

## Relation execution

RL:S:LI17:location:00:relative-to | op=directional\_relation | semantic=['LM:S:LI17:location:97-143'] | executable=['LM:S:LI17:location:101-108'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:LI17:location:01:same-level | op=same\_level\_plane | semantic=['LM:S:LI17:location:57-78'] | executable=['LM:S:LI17:location:57-78'] | dir=None | status=RESOLVED | reason=None

RL:S:LI17:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:LI17:location:7-22'] | executable=['LM:S:LI17:location:7-22'] | dir=None | status=RESOLVED | reason=None

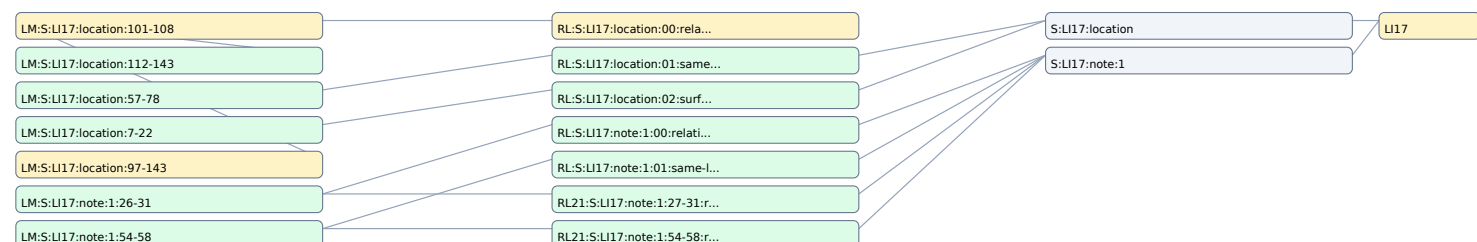
RL:S:LI17:note:1:00:relative-to | op=directional\_relation | semantic=['LM:S:LI17:note:1:26-31'] | executable=['LM:S:LI17:note:1:26-31'] | dir=inferior | status=RESOLVED | reason=None

RL:S:LI17:note:1:01:same-level | op=same\_level\_plane | semantic=['LM:S:LI17:note:1:54-58'] | executable=['LM:S:LI17:note:1:54-58'] | dir=None | status=RESOLVED | reason=None

RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 | op=reference\_dependency | semantic=['LM:S:LI17:note:1:26-31'] | executable=['LM:S:LI17:note:1:26-31'] | dir=None | status=RESOLVED | reason=None

RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e | op=reference\_dependency | semantic=['LM:S:LI17:note:1:54-58'] | executable=['LM:S:LI17:note:1:54-58'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:LI17:location:97-143 -> LM:S:LI17:location:101-108 [semantic\_to\_executable]  
LM:S:LI17:location:101-108 -> RL:S:LI17:location:00:relative-to [executable\_to\_relation]  
LM:S:LI17:location:57-78 -> RL:S:LI17:location:01:same-level [executable\_to\_relation]  
RL:S:LI17:location:01:same-level -> S:LI17:location [relation\_to\_statement]  
LM:S:LI17:location:7-22 -> RL:S:LI17:location:02:surface-landmark [executable\_to\_relation]  
RL:S:LI17:location:02:surface-landmark -> S:LI17:location [relation\_to\_statement]  
LM:S:LI17:note:1:26-31 -> RL:S:LI17:note:1:00:relative-to [executable\_to\_relation]  
RL:S:LI17:note:1:00:relative-to -> S:LI17:note:1 [relation\_to\_statement]  
LM:S:LI17:note:1:54-58 -> RL:S:LI17:note:1:01:same-level [executable\_to\_relation]  
RL:S:LI17:note:1:01:same-level -> S:LI17:note:1 [relation\_to\_statement]  
LM:S:LI17:note:1:26-31 -> RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 [executable\_to\_relation]  
RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 -> S:LI17:note:1 [relation\_to\_statement]  
LM:S:LI17:note:1:54-58 -> RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e [executable\_to\_relation]  
RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e -> S:LI17:note:1 [relation\_to\_statement]  
LM:S:LI17:location:112-143 -> LM:S:LI17:location:101-108 [parent\_to\_child\_subfeature]  
S:LI17:location -> LI17 [statement\_to\_point]  
S:LI17:note:1 -> LI17 [statement\_to\_point]

## BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:LI17:location:97-143"],"branch\_id":null,"direction":"posterior","op":"directional\_relation","source\_statement\_id":"S:LI17:location"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:location:57-78"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:LI17:location"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:location:7-22"],"branch\_id":null,"op":"surface\_membership","source\_statement\_id":"S:LI17:location"},"executor":"surface\_membership","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"branch\_id":null,"op":"reference\_dependency","source\_statement\_id":"S:LI17:note:1"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:LI17:note:1"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"branch\_id":null,"op":"reference\_dependency","source\_statement\_id":"S:LI17:note:1"},"executor":"reference\_dependency","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"branch\_id":null,"op":"reference\_dependency","source\_statement\_id":"S:LI17:note:1"},"executor":"reference\_dependency","status":"RESOLVED"}],"status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"executable\_argument\_node\_ids":["LM:S:LI17:location:101-108"],"executor":"directional\_relation","operand\_binding\_trace":[{"binding":"child\_subfeature","executable\_operand\_id":"LM:S:LI17:location:101-108","semantic\_operand\_id":"LM:S:LI17:location:97-143"}],"reason":"executable\_operand not uniquely executable","semantic\_argument\_node\_ids":["LM:S:LI17:location:97-143"],"semantic\_fields":{"branch\_id":null,"direction":"posterior","op":"directional\_relation","relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:LI17:location:97-143"],"source\_statement\_id":"S:LI17:location"},"semantic\_fields\_hash":"2b46cdf0ec4aed6638511250c33f249c16549f116b8fe76e295fa242c62fa3c","status":"UNRESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:location:57-78"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:LI17:location:57-78"],"op":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:location:57-78","semantic\_operand\_id":"LM:S:LI17:location:57-78"}],"relation\_type":"same\_level","semantic\_argument\_node\_ids":["LM:S:LI17:location:57-78"],"source\_statement\_id":"S:LI17:location"},"executable\_argument\_node\_ids":["LM:S:LI17:location:57-78"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:location:57-78","semantic\_operand\_id":"LM:S:LI17:location:57-78"}],"semantic\_argument\_node\_ids":["LM:S:LI17:location:57-78"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:LI17:location:57-78"],"source\_statement\_id":"S:LI17:location"},"semantic\_fields\_hash":"2079891f51e10f9b4f3fccf742348b1b8e6547703acac881f6d6b49cf16303a1","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:location:7-22"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:LI17:location:7-22"],"op":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:location:7-22","semantic\_operand\_id":"LM:S:LI17:location:7-22"}],"relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:LI17:location:7-22"],"semantic\_fields":{"branch\_id":null,"op":"surface\_membership","relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:LI17:location:7-22"],"source\_statement\_id":"S:LI17:location"},"semantic\_fields\_hash":"e26417b42c2975d2eaf0de9ead8d6a92717a65ab89b3cc250a57db98ea9cbab6d","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"branch\_id":null,"direction":"inferior","executable\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"op":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:note:1:26-31","semantic\_operand\_id":"LM:S:LI17:note:1:26-31"}],"relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"source\_statement\_id":"S:LI17:note:1"},"executable\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"executor":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:note:1:26-31","semantic\_operand\_id":"LM:S:LI17:note:1:26-31"}],"semantic\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"semantic\_fields":{"branch\_id":null,"direction":"inferior","op":"directional\_relation","relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"source\_statement\_id":"S:LI17:note:1"},"semantic\_fields\_hash":"a0be66cab72cc933663cc6a7a68c24cb71bd934a89a55cc053c7dbf5107b0a4","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"op":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:note:1:54-58","semantic\_operand\_id":"LM:S:LI17:note:1:54-58"}],"relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"source\_statement\_id":"S:LI17:note:1"},"executable\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:note:1:54-58","semantic\_operand\_id":"LM:S:LI17:note:1:54-58"}],"semantic\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"source\_statement\_id":"S:LI17:note:1"},"semantic\_fields\_hash":"336081d4f0f72ca176269d0db3882c4379a0046bbb843974a8718f56190029ba","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"op":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:note:1:26-31","semantic\_operand\_id":"LM:S:LI17:note:1:26-31"}],"relation\_type":"reference-acupoint","semantic\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"source\_statement\_id":"S:LI17:note:1"},"executable\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"executor":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:note:1:26-31","semantic\_operand\_id":"LM:S:LI17:note:1:26-31"}],"semantic\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"semantic\_fields":{"branch\_id":null,"op":"reference\_dependency","relation\_type":"reference-acupoint","semantic\_argument\_node\_ids":["LM:S:LI17:note:1:26-31"],"source\_statement\_id":"S:LI17:note:1"},"semantic\_fields\_hash":"d629e44ef47512180deef56a078c97af89682ce90c1051cd2d6635be6bbd35f","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"op":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:note:1:54-58","semantic\_operand\_id":"LM:S:LI17:note:1:54-58"}],"relation\_type":"reference-acupoint","semantic\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"source\_statement\_id":"S:LI17:note:1"},"executable\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"executor":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:LI17:note:1:54-58","semantic\_operand\_id":"LM:S:LI17:note:1:54-58"}],"semantic\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"semantic\_fields":{"branch\_id":null,"op":"reference\_dependency","relation\_type":"reference-acupoint","semantic\_argument\_node\_ids":["LM:S:LI17:note:1:54-58"],"source\_statement\_id":"S:LI17:note:1"},"semantic\_fields\_hash":"86d6974447c1245ad89ce39395e645deae722e8704298836571443fccf132ce","status":"RESOLVED"}],"status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

[ ] WHO source phrase 何人何語  
[ ] semantic granularity 意味  
[ ] child subfeature 子特徴 execution 実行  
[ ] distinct subfeatures 異なる実行可能 identity 同一性  
[ ] relation operator 関係演算器 operand 演算子  
[ ] condition state 条件状態  
[ ] RESOLVED/UNRESOLVED 解決済み/未解決  
[ ] ACCEPT  
[ ] REJECT

# BL17: RESOLVED -> UNRESOLVED

Defect family: modifier\_dependency\_loss

## WHO source

location | S:BL17:location | page 117 | In the upper back region, at the same level as the inferior border of the spinous process of the seventh thoracic vertebra (T7), 1.5 B-cun lateral to the posterior median line.

note | S:BL17:note:1 | page 117 | Note: The inferior angle of the scapula is at the same level as the spinous process of the seventh thoracic vertebra.

## Semantic decomposition

LM:S:BL17:location:149-175 | source= the posterior median line | class=line.anatomical\_line | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:posterior\_median\_line","subfeature\_type":null} | reason=None

LM:S:BL17:location:47-127 | source=the inferior border of the spinous process of the seventh thoracic vertebra (T7) | class=bone.bone | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL17:location:51-66 | source=inferior border | class=boundary.border | parent=LM:S:BL17:location:70-127 | child=inferior border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no approved executable construction rule for subfeature 'inferior border'

LM:S:BL17:location:7-24 | source=upper back region | class=region.body\_region | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"86974891de98cccd457e268703bc6a672fce4777ca5d0776994671dd7f004323","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:upper\_back","subfeature\_type":null} | reason=None

LM:S:BL17:location:70-127 | source=the spinous process of the seventh thoracic vertebra (T7) | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA9968","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:BL17:note:1:64-116 | source=the spinous process of the seventh thoracic vertebra | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA9968","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:BL17:note:1:75-83 | source= process | class=bone.process\_or\_prominence | parent=LM:S:BL17:note:1:87-116 | child= process | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=child subfeature type not recognized

LM:S:BL17:note:1:87-116 | source=the seventh thoracic vertebra | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA9968","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

## Composite binding view

PARENT LM:S:BL17:location:70-127 the spinous process of the seventh thoracic vertebra (T7) status=RESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA9968","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:BL17:location:51-66 inferior border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

PARENT LM:S:BL17:note:1:87-116 the seventh thoracic vertebra status=RESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA9968","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:BL17:note:1:75-83 process status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

## Relation execution

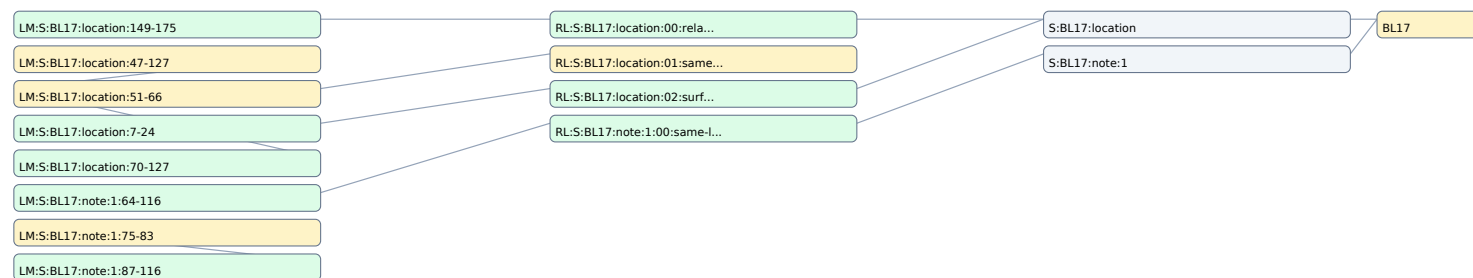
RL:S:BL17:location:00:relative-to | op=directional\_relation | semantic=['LM:S:BL17:location:149-175'] | executable=['LM:S:BL17:location:149-175'] | dir=lateral | status=RESOLVED | reason=None

RL:S:BL17:location:01:same-level | op=same\_level\_plane | semantic=['LM:S:BL17:location:47-127'] | executable=['LM:S:BL17:location:51-66'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:BL17:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:BL17:location:7-24'] | executable=['LM:S:BL17:location:7-24'] | dir=None | status=RESOLVED | reason=None

RL:S:BL17:note:1:00:same-level | op=same\_level\_plane | semantic=['LM:S:BL17:note:1:64-116'] | executable=['LM:S:BL17:note:1:64-116'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:BL17:location:149-175 -> RL:S:BL17:location:00:relative-to [executable\_to\_relation]  
 RL:S:BL17:location:00:relative-to -> S:BL17:location [relation\_to\_statement]  
 LM:S:BL17:location:47-127 -> LM:S:BL17:location:51-66 [semantic\_to\_executable]  
 LM:S:BL17:location:51-66 -> RL:S:BL17:location:01:same-level [executable\_to\_relation]  
 LM:S:BL17:location:7-24 -> RL:S:BL17:location:02:surface-landmark [executable\_to\_relation]  
 RL:S:BL17:location:02:surface-landmark -> S:BL17:location [relation\_to\_statement]  
 LM:S:BL17:note:1:64-116 -> RL:S:BL17:note:1:00:same-level [executable\_to\_relation]  
 RL:S:BL17:note:1:00:same-level -> S:BL17:note:1 [relation\_to\_statement]  
 LM:S:BL17:location:70-127 -> LM:S:BL17:location:51-66 [parent\_to\_child\_subfeature]  
 LM:S:BL17:note:1:87-116 -> LM:S:BL17:note:1:75-83 [parent\_to\_child\_subfeature]  
 S:BL17:location -> BL17 [statement\_to\_point]  
 S:BL17:note:1 -> BL17 [statement\_to\_point]

## BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:BL17:location:149-175"],"branch\_id":null,"direction":"lateral","op":"directional\_relation","source\_statement\_id":"S:BL17:location"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL17:location:47-127"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:BL17:location"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL17:location:7-24"],"branch\_id":null,"op":"surface\_membership","source\_statement\_id":"S:BL17:location"},"executor":"surface\_membership","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL17:note:1:64-116"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:BL17:note:1"},"executor":"same\_level\_plane","status":"RESOLVED"}], "status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:BL17:location:149-175"],"branch\_id":null,"direction":"lateral","executable\_argument\_node\_ids":["LM:S:BL17:location:149-175"],"op":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL17:location:149-175"},"semantic\_operand\_id":"LM:S:BL17:location:149-175"],"relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:BL17:location:149-175"],"source\_statement\_id":"S:BL17:location"},"executable\_argument\_node\_ids":["LM:S:BL17:location:149-175"],"executor":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL17:location:149-175"},"semantic\_operand\_id":"LM:S:BL17:location:149-175"],"semantic\_argument\_node\_ids":["LM:S:BL17:location:149-175"],"semantic\_fields":{"branch\_id":null,"direction":"lateral","op":"directional\_relation","relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:BL17:location:149-175"],"source\_statement\_id":"S:BL17:location"},"semantic\_fields\_hash":"6a11aebbd715e5ef3e3587f2e1aa7b7282d788fa5b565c356652258e3dbe2d83","status":"RESOLVED"}, {"executable\_argument\_node\_ids":["LM:S:BL17:location:51-66"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"child\_subfeature","executable\_operand\_id":"LM:S:BL17:location:51-66"},"semantic\_operand\_id":"LM:S:BL17:location:47-127"}], "reason":"executable operand not uniquely executable","semantic\_argument\_node\_ids":["LM:S:BL17:location:47-127"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:BL17:location:47-127"],"source\_statement\_id":"S:BL17:location"},"semantic\_fields\_hash":"ba54059035ed65df89be107cec93ea882cff2412857d6999e48ba9913b696be0","status":"UNRESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL17:location:7-24"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:BL17:location:7-24"],"op":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL17:location:7-24"},"semantic\_operand\_id":"LM:S:BL17:location:7-24"}], "relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:BL17:location:7-24"],"source\_statement\_id":"S:BL17:location"},"executable\_argument\_node\_ids":["LM:S:BL17:location:7-24"],"executor":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL17:location:7-24"},"semantic\_operand\_id":"LM:S:BL17:location:7-24"},"semantic\_argument\_node\_ids":["LM:S:BL17:location:7-24"],"semantic\_fields":{"branch\_id":null,"op":"surface\_membership","relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:BL17:location:7-24"],"source\_statement\_id":"S:BL17:location"},"semantic\_fields\_hash":"d6e9916331460e32751b849ed1c04c2e73fee2c4438a77ea1f7d9f1c787d74f7","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL17:note:1:64-116"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:BL17:note:1:64-116"],"op":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL17:note:1:64-116"},"semantic\_operand\_id":"LM:S:BL17:note:1:64-116"}], "relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:BL17:note:1:64-116"],"source\_statement\_id":"S:BL17:note:1"},"executable\_argument\_node\_ids":["LM:S:BL17:note:1:64-116"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL17:note:1:64-116"},"semantic\_operand\_id":"LM:S:BL17:note:1:64-116"}], "semantic\_argument\_node\_ids":["LM:S:BL17:note:1:64-116"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:BL17:note:1:64-116"],"source\_statement\_id":"S:BL17:note:1"},"semantic\_fields\_hash":"bb6a8a9dc0a5e2bdeaade9a8bbe219bb21a459878612bbec0e7d0d1180ce959c","status":"RESOLVED"}], "status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

[ ] WHO source phrase [ ] [ ] [ ]  
 [ ] semantic granularity [ ] [ ]  
 [ ] child subfeature [ ] [ ] execution [ ] [ ]  
 [ ] distinct subfeatures [ ] distinct executable identity [ ] [ ]  
 [ ] relation operator [ ] [ ] operand [ ] [ ]  
 [ ] condition state [ ] [ ] [ ] [ ]  
 [ ] RESOLVED/UNRESOLVED [ ] [ ] [ ] [ ]  
 [ ] ACCEPT  
 [ ] REJECT

# BL23: RESOLVED -> UNRESOLVED

Defect family: modifier\_dependency\_loss

## WHO source

location | S:BL23:location | page 120 | In the lumbar region, at the same level as the inferior border of the spinous process of the second lumbar vertebra (L2), 1.5 B-cun lateral to the posterior median line.

## Semantic decomposition

LM:S:BL23:location:142-168 | source= the posterior median line | class=line.anatomical\_line | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:posterior\_median\_line","subfeature\_type":null} | reason=None

LM:S:BL23:location:43-120 | source=the inferior border of the spinous process of the second lumbar vertebra (L2) | class=bone.bone | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL23:location:47-62 | source=inferior border | class=boundary.border | parent=LM:S:BL23:location:66-120 | child=inferior border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no approved executable construction rule for subfeature 'inferior border'

LM:S:BL23:location:66-120 | source=the spinous process of the second lumbar vertebra (L2) | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13073","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:BL23:location:7-20 | source=lumbar region | class=region.body\_region | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"ef7e2ce29e7bdfb2606e98c7c2254ff2a0c99337bcf65192272b78f367ce7244","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:lumbar\_region","subfeature\_type":null} | reason=None

## Composite binding view

PARENT LM:S:BL23:location:66-120 the spinous process of the second lumbar vertebra (L2) status=RESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13073","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:BL23:location:47-62 inferior border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

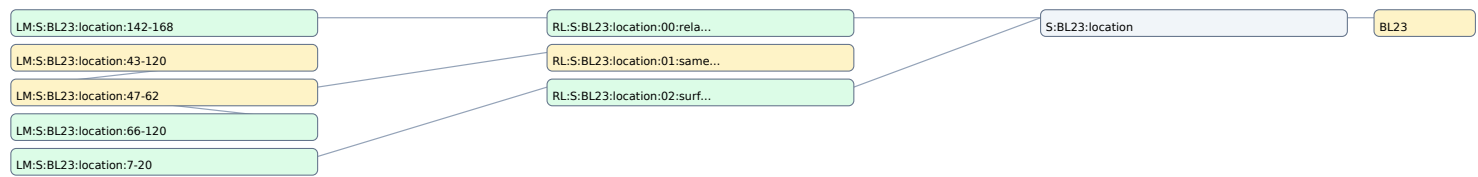
## Relation execution

RL:S:BL23:location:00:relative-to | op=directional\_relation | semantic=['LM:S:BL23:location:142-168'] | executable=['LM:S:BL23:location:142-168'] | dir=lateral | status=RESOLVED | reason=None

RL:S:BL23:location:01:same-level | op=same\_level\_plane | semantic=['LM:S:BL23:location:43-120'] | executable=['LM:S:BL23:location:47-62'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:BL23:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:BL23:location:7-20'] | executable=['LM:S:BL23:location:7-20'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

- LM:S:BL23:location:142-168 -> RL:S:BL23:location:00:relative-to [executable\_to\_relation]
- RL:S:BL23:location:00:relative-to -> S:BL23:location [relation\_to\_statement]
- LM:S:BL23:location:43-120 -> LM:S:BL23:location:47-62 [semantic\_to\_executable]
- LM:S:BL23:location:47-62 -> RL:S:BL23:location:01:same-level [executable\_to\_relation]
- LM:S:BL23:location:7-20 -> RL:S:BL23:location:02:surface-landmark [executable\_to\_relation]
- RL:S:BL23:location:02:surface-landmark -> S:BL23:location [relation\_to\_statement]
- LM:S:BL23:location:66-120 -> LM:S:BL23:location:47-62 [parent\_to\_child\_subfeature]
- S:BL23:location -> BL23 [statement\_to\_point]

## BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:BL23:location:142-168"],"branch\_id":null,"direction":"lateral","op":"directional\_relation","source\_statement\_id":"S:BL23:location"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL23:location:43-120"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:BL23:location"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL23:location:7-20"],"branch\_id":null,"op":"surface\_membership","source\_statement\_id":"S:BL23:location"},"executor":"surface\_membership","status":"RESOLVED"}], "status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:BL23:location:142-168"],"branch\_id":null,"direction":"lateral","executable\_argument\_node\_ids":["LM:S:BL23:location:142-168"],"op":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL23:location:142-168","semantic\_operand\_id":"LM:S:BL23:location:142-168"}],"relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:BL23:location:142-168"],"source\_statement\_id":"S:BL23:location"},"executable\_argument\_node\_ids":["LM:S:BL23:location:142-168"],"executor":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL23:location:142-168","semantic\_operand\_id":"LM:S:BL23:location:142-168"}],"semantic\_argument\_node\_ids":["LM:S:BL23:location:142-168"],"semantic\_fields":{"branch\_id":null,"direction":"lateral","op":"directional\_relation","relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:BL23:location:142-168"],"source\_statement\_id":"S:BL23:location"},"semantic\_fields\_hash":"862c78830cd2dc16693d8cbc4a56f30c9ce3fa1488debbbc6fa903edfc04af189","status":"RESOLVED"},{"executable\_argument\_node\_ids":["LM:S:BL23:location:47-62"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"child\_subfeature","executable\_operand\_id":"LM:S:BL23:location:47-62","semantic\_operand\_id":"LM:S:BL23:location:43-120"}],"reason":"executable operand not uniquely executable","semantic\_argument\_node\_ids":["LM:S:BL23:location:43-120"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:BL23:location:43-120"],"source\_statement\_id":"S:BL23:location"},"semantic\_fields\_hash":"88e3397ffd20831a035aef5ee673749ac0196234b6b92198de2a59f96ab3b7d5","status":"UNRESOLVED"},{"constraint":{"argument\_node\_ids":["LM:S:BL23:location:7-20"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:BL23:location:7-20"],"op":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL23:location:7-20","semantic\_operand\_id":"LM:S:BL23:location:7-20"}],"relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:BL23:location:7-20"],"source\_statement\_id":"S:BL23:location"},"executable\_argument\_node\_ids":["LM:S:BL23:location:7-20"],"executor":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL23:location:7-20","semantic\_operand\_id":"LM:S:BL23:location:7-20"}],"semantic\_argument\_node\_ids":["LM:S:BL23:location:7-20"],"semantic\_fields":{"branch\_id":null,"op":"surface\_membership","relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:BL23:location:7-20"],"source\_statement\_id":"S:BL23:location"},"semantic\_fields\_hash":"7ef1f11395cf9bf57fb05cd3712f0ec2325b6f4c4c713ed8ed29a82e2bdb0323","status":"RESOLVED"}],"status":"UNRESOLVED"}]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase 词性 词性
- [ ] semantic granularity 词性
- [ ] child subfeature 词性 词性 execution 词性
- [ ] distinct subfeatures 词性 distinct executable identity 词性
- [ ] relation operator 词性 词性 operand 词性
- [ ] condition state 词性 词性 词性
- [ ] RESOLVED/UNRESOLVED 词性 词性
- [ ] ACCEPT
- [ ] REJECT

# BL25: RESOLVED -> UNRESOLVED

Defect family: modifier\_dependency\_loss

## WHO source

location | S:BL25:location | page 121 | In the lumbar region, at the same level as the inferior border of the spinous process of the fourth lumbar vertebra (L4), 1.5 B-cun lateral to the posterior median line.

## Semantic decomposition

LM:S:BL25:location:142-168 | source= the posterior median line | class=line.anatomical\_line | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:posterior\_median\_line","subfeature\_type":null} | reason=None

LM:S:BL25:location:43-120 | source=the inferior border of the spinous process of the fourth lumbar vertebra (L4) | class=bone.bone | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:BL25:location:47-62 | source=inferior border | class=boundary.border | parent=LM:S:BL25:location:66-120 | child=inferior border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no approved executable construction rule for subfeature 'inferior border'

LM:S:BL25:location:66-120 | source=the spinous process of the fourth lumbar vertebra (L4) | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13075","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:BL25:location:7-20 | source=lumbar region | class=region.body\_region | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"ef7e2ce29e7bdfb2606e98c7c2254ff2a0c99337bcf65192272b78f367ce7244","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:lumbar\_region","subfeature\_type":null} | reason=None

## Composite binding view

PARENT LM:S:BL25:location:66-120 the spinous process of the fourth lumbar vertebra (L4) status=RESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13075","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:BL25:location:47-62 inferior border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

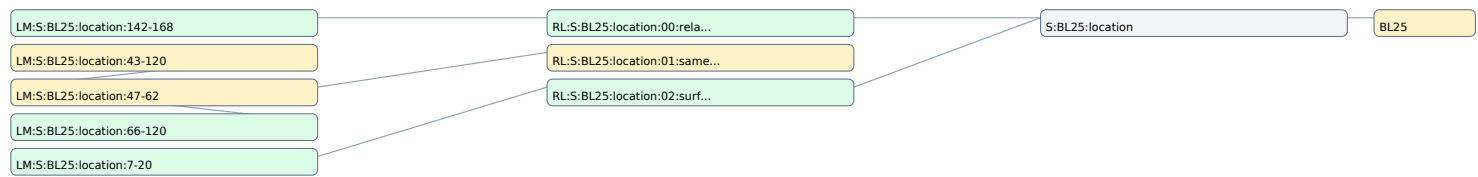
## Relation execution

RL:S:BL25:location:00:relative-to | op=directional\_relation | semantic=['LM:S:BL25:location:142-168'] | executable=['LM:S:BL25:location:142-168'] | dir=lateral | status=RESOLVED | reason=None

RL:S:BL25:location:01:same-level | op=same\_level\_plane | semantic=['LM:S:BL25:location:43-120'] | executable=['LM:S:BL25:location:47-62'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:BL25:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:BL25:location:7-20'] | executable=['LM:S:BL25:location:7-20'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

- LM:S:BL25:location:142-168 -> RL:S:BL25:location:00:relative-to [executable\_to\_relation]
- RL:S:BL25:location:00:relative-to -> S:BL25:location [relation\_to\_statement]
- LM:S:BL25:location:43-120 -> LM:S:BL25:location:47-62 [semantic\_to\_executable]
- LM:S:BL25:location:47-62 -> RL:S:BL25:location:01:same-level [executable\_to\_relation]
- LM:S:BL25:location:7-20 -> RL:S:BL25:location:02:surface-landmark [executable\_to\_relation]
- RL:S:BL25:location:02:surface-landmark -> S:BL25:location [relation\_to\_statement]
- LM:S:BL25:location:66-120 -> LM:S:BL25:location:47-62 [parent\_to\_child\_subfeature]
- S:BL25:location -> BL25 [statement\_to\_point]

## BEFORE / AFTER diff

BEFORE: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:BL25:location:142-168"],"branch\_id":null,"direction":"lateral","op":"directional\_relation","source\_statement\_id":"S:BL25:location"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL25:location:43-120"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:BL25:location"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:BL25:location:7-20"],"branch\_id":null,"op":"surface\_membership","source\_statement\_id":"S:BL25:location"},"executor":"surface\_membership","status":"RESOLVED"}], "status":"RESOLVED"}

AFTER: {"conditions":[],"relations":[{"constraint":{"argument\_node\_ids":["LM:S:BL25:location:142-168"],"branch\_id":null,"direction":"lateral","executable\_argument\_node\_ids":["LM:S:BL25:location:142-168"],"op":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL25:location:142-168","semantic\_operand\_id":"LM:S:BL25:location:142-168"}],"relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:BL25:location:142-168"],"source\_statement\_id":"S:BL25:location"},"executable\_argument\_node\_ids":["LM:S:BL25:location:142-168"],"executor":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL25:location:142-168","semantic\_operand\_id":"LM:S:BL25:location:142-168"}],"semantic\_argument\_node\_ids":["LM:S:BL25:location:142-168"],"semantic\_fields":{"branch\_id":null,"direction":"lateral","op":"directional\_relation","relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:BL25:location:142-168"],"source\_statement\_id":"S:BL25:location"},"semantic\_fields\_hash":"1b7ca304871e9a64afc4612124bd256897d067a7d065c8e7d99a16801ffc811e","status":"RESOLVED"},"executable\_argument\_node\_ids":["LM:S:BL25:location:47-62"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"child\_subfeature","executable\_operand\_id":"LM:S:BL25:location:47-62","semantic\_operand\_id":"LM:S:BL25:location:43-120"}],"reason":"executable operand not uniquely executable","semantic\_argument\_node\_ids":["LM:S:BL25:location:43-120"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:BL25:location:43-120"],"source\_statement\_id":"S:BL25:location"},"semantic\_fields\_hash":"caa9e7786c7095e3489c3ea24114d5e5c8e105a2b8423ad52855af499ac4adbe","status":"UNRESOLVED"},"constraint":{"argument\_node\_ids":["LM:S:BL25:location:7-20"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:BL25:location:7-20"],"op":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL25:location:7-20","semantic\_operand\_id":"LM:S:BL25:location:7-20"}],"relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:BL25:location:7-20"],"source\_statement\_id":"S:BL25:location"},"executable\_argument\_node\_ids":["LM:S:BL25:location:7-20"],"executor":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:BL25:location:7-20","semantic\_operand\_id":"LM:S:BL25:location:7-20"}],"semantic\_argument\_node\_ids":["LM:S:BL25:location:7-20"],"semantic\_fields":{"branch\_id":null,"op":"surface\_membership","relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:BL25:location:7-20"],"source\_statement\_id":"S:BL25:location"},"semantic\_fields\_hash":"53f2ca5e3ce9d576a596085b32ffa1b87d88cda84bb7d347526593f3e50332fc","status":"RESOLVED"},"status":"UNRESOLVED"}

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase 0000 0000
- [ ] semantic granularity 0000
- [ ] child subfeature 00 execution 0000
- [ ] distinct subfeatures 00 distinct executable identity 00 00
- [ ] relation operator 0000 operand 0000
- [ ] condition state 0000 0000
- [ ] RESOLVED/UNRESOLVED 0000 0000
- [ ] ACCEPT
- [ ] REJECT

# ST9: RESOLVED -> UNRESOLVED

Defect family: modifier\_dependency\_loss

## WHO source

location | S:ST9:location | page 59 | In the anterior region of the neck, at the same level as the superior border of the thyroid cartilage, anterior to the sterno-cleidomastoid muscle, over the common carotid artery.

note | S:ST9:note:1 | page 59 | Note 1: The sternocleidomastoid muscle is more distinct when the head is turned to the opposite side against resistance.

note | S:ST9:note:2 | page 59 | Note 2: ST9 is located at the same level as LI18, SI16 and the superior border of the thyroid cartilage. St9 is located anterior to the sternocleidomastoid muscle, and SI16, posterior to the sternocleidomastoid muscle and LI18, between the anterior and pos-terior borders of the sternocleidomastoid muscle.

## Semantic decomposition

LM:S:ST9:location:114-146 | source= the sterno-cleidomastoid muscle | class=muscle | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13407","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST9:location:157-178 | source=common carotid artery | class=vessel | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA3939","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST9:location:57-101 | source=the superior border of the thyroid cartilage | class=cartilage | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST9:location:61-76 | source=superior border | class=boundary.border | parent=LM:S:ST9:location:84-101 | child=superior border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no approved executable construction rule for subfeature 'superior border'

LM:S:ST9:location:7-34 | source=anterior region of the neck | class=region.body\_region | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"71f7a5ab627b3c93a9aceb574aabdb27ca3577688316b7de105a5effa7840dc7","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:anterior\_neck","subfeature\_type":null} | reason=None

LM:S:ST9:location:84-101 | source=thyroid cartilage | class=cartilage | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA55099","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST9:note:2:131-162 | source= the sternocleidomastoid muscle | class=muscle | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13407","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST9:note:2:186-226 | source= the sternocleidomastoid muscle and LI18 | class=muscle | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA13407","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST9:note:2:240-305:6f7a364e | source=anterior and pos-terior borders of the sternocleidomastoid muscle | class=muscle | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST9:note:2:253-305:a442a1a4 | source=pos-terior borders of the sternocleidomastoid muscle | class=muscle | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST9:note:2:44-48 | source=LI18 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"LI18","registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST9:note:2:63-78 | source=superior border | class=boundary.border | parent=LM:S:ST9:note:2:82-103 | child=superior border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no approved executable construction rule for subfeature 'superior border'

LM:S:ST9:note:2:82-103 | source=the thyroid cartilage | class=cartilage | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA55099","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM21:S:ST9:note:2:50-54:6775aaafef | source=SI16 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"SI16","registry\_id":null,"subfeature\_type":null} | reason=None

LM21:S:ST9:note:2:168-172:6775aaafef | source=SI16 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"SI16","registry\_id":null,"subfeature\_type":null} | reason=None

LM21:S:ST9:note:2:222-226:34574dfae8 | source=LI18 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"LI18","registry\_id":null,"subfeature\_type":null} | reason=None

## Composite binding view

PARENT LM:S:ST9:location:84-101 thyroid cartilage status=RESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA55099","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:ST9:location:61-76 superior border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

PARENT LM:S:ST9:note:2:82-103 the thyroid cartilage status=RESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA55099","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:ST9:note:2:63-78 superior border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

Relation execution

RL:S:ST9:location:00:relative-to | op=directional\_relation | semantic=['LM:S:ST9:location:114-146'] | executable=['LM:S:ST9:location:114-146'] | dir=anterior | status=RESOLVED | reason=None

RL:S:ST9:location:01:same-level | op=same\_level\_plane | semantic=['LM:S:ST9:location:57-101'] | executable=['LM:S:ST9:location:61-76'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST9:location:02:overlies | op=overlies\_projection | semantic=['LM:S:ST9:location:157-178'] | executable=['LM:S:ST9:location:157-178'] | dir=None | status=RESOLVED | reason=None

RL:S:ST9:location:03:surface-landmark | op=surface\_membership | semantic=['LM:S:ST9:location:7-34'] | executable=['LM:S:ST9:location:7-34'] | dir=None | status=RESOLVED | reason=None

RL:S:ST9:note:2:00:between | op=between\_constraint | semantic=['LM:S:ST9:note:2:240-305:6f7a364e', 'LM:S:ST9:note:2:253-305:a442a1a4'] | executable=[] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:ST9:note:2:01:relative-to | op=directional\_relation | semantic=['LM:S:ST9:note:2:131-162'] | executable=['LM:S:ST9:note:2:131-162'] | dir=anterior | status=RESOLVED | reason=None

RL:S:ST9:note:2:02:relative-to | op=directional\_relation | semantic=['LM:S:ST9:note:2:186-226'] | executable=['LM:S:ST9:note:2:186-226'] | dir=posterior | status=RESOLVED | reason=None

RL:S:ST9:note:2:03:same-level | op=same\_level\_plane | semantic=['LM:S:ST9:note:2:44-48'] | executable=['LM:S:ST9:note:2:44-48'] | dir=None | status=RESOLVED | reason=None

RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 | op=reference\_dependency | semantic=['LM:S:ST9:note:2:44-48'] | executable=['LM:S:ST9:note:2:44-48'] | dir=None | status=RESOLVED | reason=None

RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c | op=reference\_dependency | semantic=['LM21:S:ST9:note:2:50-54:6775aaafef'] | executable=['LM21:S:ST9:note:2:50-54:6775aaafef'] | dir=None | status=RESOLVED | reason=None

RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 | op=reference\_dependency | semantic=['LM21:S:ST9:note:2:168-172:6775aaafef'] | executable=['LM21:S:ST9:note:2:168-172:6775aaafef'] | dir=None | status=RESOLVED | reason=None

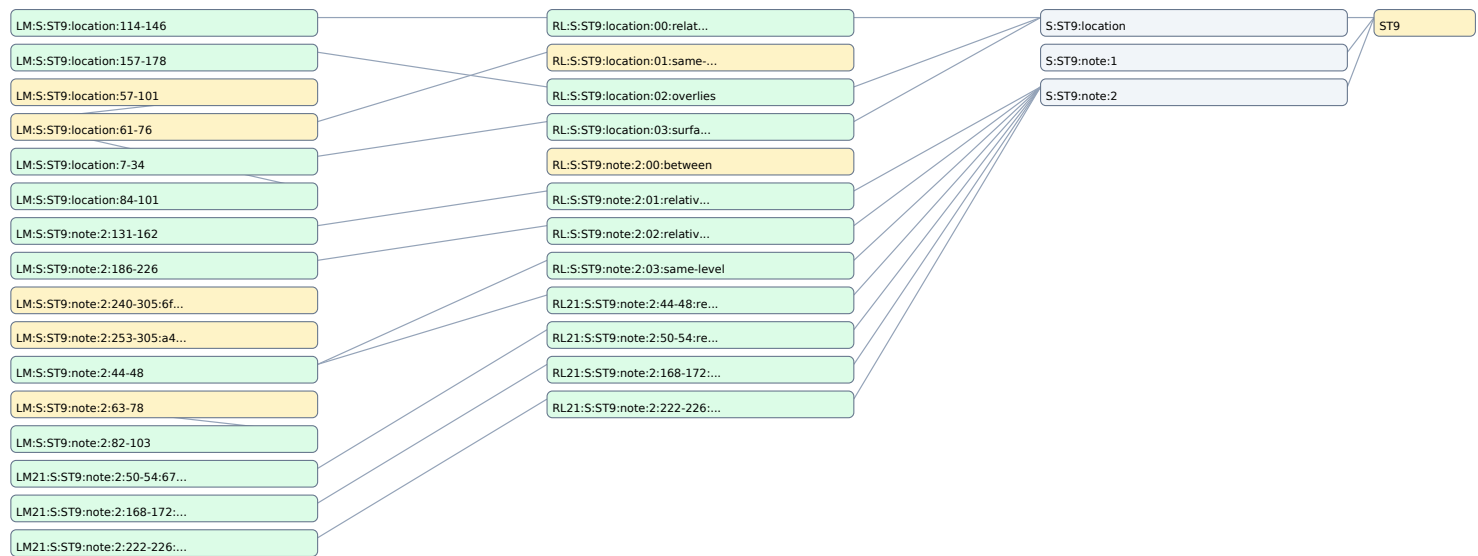
RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 | op=reference\_dependency | semantic=['LM21:S:ST9:note:2:222-226:34574dfae8'] | executable=['LM21:S:ST9:note:2:222-226:34574dfae8'] | dir=None | status=RESOLVED | reason=None

Conditions

{"condition\_id":"CD:S:ST9:note:1:00","condition\_type":"body\_position","default\_pose\_compatible":false,"reason":"condition requires pose incompatible with default BodyParts3D pose","repaired\_status":"CONDITIONAL","source\_raw":"when the head is turned to the opposite side against resistance","source\_statement\_id":"S:ST9:note:1"}

{"condition\_id":"CD:S:ST9:note:1:01","condition\_type":"palpation\_dependent","default\_pose\_compatible":null,"reason":"condition preserved as non-executed conditional branch","repaired\_status":"CONDITIONAL","source\_raw":"more distinct when the head is turned to the opposite side against resistance","source\_statement\_id":"S:ST9:note:1"}

Dependency graph



Dependency graph edge list

LM:S:ST9:location:114-146 -> RL:S:ST9:location:00:relative-to [executable\_to\_relation]

RL:S:ST9:location:00:relative-to -> S:ST9:location [relation\_to\_statement]

LM:S:ST9:location:57-101 -> LM:S:ST9:location:61-76 [semantic\_to\_executable]  
LM:S:ST9:location:61-76 -> RL:S:ST9:location:01:same-level [executable\_to\_relation]  
LM:S:ST9:location:157-178 -> RL:S:ST9:location:02:overlies [executable\_to\_relation]  
RL:S:ST9:location:02:overlies -> S:ST9:location [relation\_to\_statement]  
LM:S:ST9:location:7-34 -> RL:S:ST9:location:03:surface-landmark [executable\_to\_relation]  
RL:S:ST9:location:03:surface-landmark -> S:ST9:location [relation\_to\_statement]  
LM:S:ST9:note:2:131-162 -> RL:S:ST9:note:2:01:relative-to [executable\_to\_relation]  
RL:S:ST9:note:2:01:relative-to -> S:ST9:note:2 [relation\_to\_statement]  
LM:S:ST9:note:2:186-226 -> RL:S:ST9:note:2:02:relative-to [executable\_to\_relation]  
RL:S:ST9:note:2:02:relative-to -> S:ST9:note:2 [relation\_to\_statement]  
LM:S:ST9:note:2:44-48 -> RL:S:ST9:note:2:03:same-level [executable\_to\_relation]  
RL:S:ST9:note:2:03:same-level -> S:ST9:note:2 [relation\_to\_statement]  
LM:S:ST9:note:2:44-48 -> RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 [executable\_to\_relation]  
RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 -> S:ST9:note:2 [relation\_to\_statement]  
LM21:S:ST9:note:2:50-54:6775aaafeb -> RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c [executable\_to\_relation]  
RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c -> S:ST9:note:2 [relation\_to\_statement]  
LM21:S:ST9:note:2:168-172:6775aaafeb -> RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 [executable\_to\_relation]  
RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 -> S:ST9:note:2 [relation\_to\_statement]  
LM21:S:ST9:note:2:222-226:34574dfae8 -> RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 [executable\_to\_relation]  
RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 -> S:ST9:note:2 [relation\_to\_statement]  
LM:S:ST9:location:84-101 -> LM:S:ST9:location:61-76 [parent\_to\_child\_subfeature]  
LM:S:ST9:note:2:82-103 -> LM:S:ST9:note:2:63-78 [parent\_to\_child\_subfeature]  
S:ST9:location -> ST9 [statement\_to\_point]  
S:ST9:note:1 -> ST9 [statement\_to\_point]  
S:ST9:note:2 -> ST9 [statement\_to\_point]

## BEFORE / AFTER diff

BEFORE: {"conditions":[{"branch\_id":"branch\_1","condition\_type":"body\_position","executor":"condition\_contract","reason":"condition requires pose incompatible with default BodyParts3D pose"},"status":"CONDITIONAL"}, {"branch\_id":"branch\_2","condition\_type":"palpation\_dependent","executor":"condition\_contract","reason":"branch preserved; not forced into default-pose primary computation","status":"CONDITIONAL"}], "relations":[{"constraint":{"argument\_node\_ids":["LM:S:ST9:location:114-146"],"branch\_id":null,"direction":"anterior","op":"directional\_relation","source\_statement\_id":"S:ST9:location"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:location:57-101"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:ST9:location"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:location:157-178"],"branch\_id":null,"op":"overlies\_projection","source\_statement\_id":"S:ST9:location"},"executor":"overlies\_projection","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:location:7-34"],"branch\_id":null,"op":"surface\_membership","source\_statement\_id":"S:ST9:location"},"executor":"surface\_membership","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:240-305:6f7a364e"],"branch\_id":null,"op":"between\_constraint","source\_statement\_id":"S:ST9:note:2"},"executor":"between\_constraint","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:131-162"],"branch\_id":null,"direction":"anterior","op":"directional\_relation","source\_statement\_id":"S:ST9:note:2"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:186-226"],"branch\_id":null,"direction":"posterior","op":"directional\_relation","source\_statement\_id":"S:ST9:note:2"},"executor":"directional\_relation","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:222-226:34574dfae8"],"branch\_id":null,"op":"same\_level\_plane","source\_statement\_id":"S:ST9:note:2"},"executor":"same\_level\_plane","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"branch\_id":null,"op":"reference\_dependency","source\_statement\_id":"S:ST9:note:2"},"executor":"reference\_dependency","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM21:S:ST9:note:2:50-54:6775aaafeb"],"branch\_id":null,"op":"reference\_dependency","source\_statement\_id":"S:ST9:note:2"},"executor":"reference\_dependency","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM21:S:ST9:note:2:168-172:6775aaafeb"],"branch\_id":null,"op":"reference\_dependency","source\_statement\_id":"S:ST9:note:2"},"executor":"reference\_dependency","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"branch\_id":null,"op":"reference\_dependency","source\_statement\_id":"S:ST9:note:2"},"executor":"reference\_dependency","status":"RESOLVED"}], "status":"RESOLVED"}

AFTER: {"conditions":[{"branch\_id":"branch\_1","condition\_type":"body\_position","default\_pose\_compatible":false,"executor":"condition\_contract","reason":"condition requires pose incompatible with default BodyParts3D pose","source\_span":{"char\_end":119,"char\_start":56,"condition\_type":"body\_position","role":null,"source\_raw":"when the head is turned to the opposite side against resistance"},"source\_statement\_id":"S:ST9:note:1","status":"CONDITIONAL"}, {"branch\_id":"branch\_2","condition\_type":"palpation\_dependent","default\_pose\_compatible":null,"executor":"condition\_contract","reason":"condition preserved as non-executed conditional branch","source\_span":{"char\_end":119,"char\_start":42,"condition\_type":"palpation\_dependent","role":null,"source\_raw":"more distinct when the head is turned to the opposite side against resistance"},"source\_statement\_id":"S:ST9:note:1","status":"CONDITIONAL"}], "relations":[{"constraint":{"argument\_node\_ids":["LM:S:ST9:location:114-146"],"branch\_id":null,"direction":"anterior","executable\_argument\_node\_ids":["LM:S:ST9:location:114-146"],"op":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:location:114-146","semantic\_operand\_id":"LM:S:ST9:location:114-146"}], "relation\_type":"relative-to","semantic\_argument\_node\_ids":["LM:S:ST9:location:114-146"],"source\_statement\_id":"S:ST9:location"},"executable\_argument\_node\_ids":["LM:S:ST9:location:114-146"],"semantic\_fields\_hash":"0da925091aded9811ecc3019b8f660367a31b928ea97672e386fe0636dbcb40a9","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:location:61-76"],"branch\_id":null,"op":"same\_level\_plane","executable\_operand\_id":"LM:S:ST9:location:61-76","semantic\_operand\_id":"LM:S:ST9:location:57-101"},"reason":"executable operand not uniquely executable","semantic\_argument\_node\_ids":["LM:S:ST9:location:57-101"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level","semantic\_argument\_node\_ids":["LM:S:ST9:location:57-101"],"source\_statement\_id":"S:ST9:location"},"semantic\_fields\_hash":"00f74e5d21de535782c60963d3309c26eba9a4c782fd55d42a5d9ac524fd1aee","status":"UNRESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:location:157-178"],"branch\_id":null,"op":"overlies\_projection","executable\_operand\_id":"LM:S:ST9:location:157-178","semantic\_operand\_id":"LM:S:ST9:location:157-178"},"relation\_type":"overlies","semantic\_argument\_node\_ids":["LM:S:ST9:location:157-178"],"source\_statement\_id":"S:ST9:location"},"executable\_argument\_node\_ids":["LM:S:ST9:location:157-178"],"semantic\_fields\_hash":"a394ba9959b2e061747e70591eaf347298f1274a1db495465697d2d4c5a29194","status":"RESOLVED"}, {"constraint":{"argument\_node\_ids":["LM:S:ST9:location:7-34"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:ST9:location:7-34"],"op":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:location:7-34","semantic\_operand\_id":"LM:S:ST9:location:7-34"}], "relation\_type":"surface-landmark","semantic\_argument\_node\_ids":["LM:S:ST9:location:7-34"]}]}

tion:7-34"],"source\_statement\_id":"S:ST9:location"},"executable\_argument\_node\_ids":["LM:S:ST9:location:7-34"],"executor":"surface\_membership","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:location:7-34","semantic\_operand\_id":"LM:S:ST9:location:7-34"}],"semantic\_argument\_node\_ids":["LM:S:ST9:location:7-34"],"semantic\_fields":{"branch\_id":null,"op":"surface\_membership","relation\_type":"surface-landmark"},"semantic\_argument\_node\_ids":["LM:S:ST9:location:7-34"],"source\_statement\_id":"S:ST9:location"},"semantic\_fields\_hash":"7988af80d90df68d16d40f7455be910ddfbadf4dd610a05451eddf6182dfb4a"},"status":"RESOLVED"},"executable\_argument\_node\_ids":["LM:S:ST9:location:7-34"],"executor":"between\_constraint","operand\_binding\_trace":[{"binding":"missing\_executable\_subfeature","executable\_operand\_ids":["LM:S:ST9:note:2:240-305:6f7a364e"],"binding":"missing\_executable\_subfeature","executable\_operand\_ids":["LM:S:ST9:note:2:253-305:a442a1a4"]},"reason":"semantic composite has multiple child subfeatures and no unique executable operand"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:240-305:6f7a364e","LM:S:ST9:note:2:253-305:a442a1a4"],"semantic\_fields":{"branch\_id":null,"op":"between\_constraint","relation\_type":"between"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:240-305:6f7a364e","LM:S:ST9:note:2:253-305:a442a1a4"],"source\_statement\_id":"S:ST9:note:2"},"semantic\_fields\_hash":"0de1e68e63b08635e4dc6643430e9b8efdd9399b349d8343db4738de254e0319"},"status":"UNRESOLVED"},"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:131-162"],"branch\_id":null,"direction":"anterior","executable\_argument\_node\_ids":["LM:S:ST9:note:2:131-162"],"op":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:note:2:131-162"},"semantic\_operand\_id":"LM:S:ST9:note:2:131-162"]},"relation\_type":"relative-to"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:131-162"],"source\_statement\_id":"S:ST9:note:2"},"executable\_argument\_node\_ids":["LM:S:ST9:note:2:131-162"],"executor":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:note:2:131-162"},"semantic\_operand\_id":"LM:S:ST9:note:2:131-162"]},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:131-162"],"semantic\_fields":{"branch\_id":null,"op":"directional\_relation","relation\_type":"relative-to"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:131-162"],"source\_statement\_id":"S:ST9:note:2"},"semantic\_fields\_hash":"3855ed87478f75dc47705e1eee3b92e08423a7e178d215c99ce3c0fde874c76a"},"status":"RESOLVED"},"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:186-226"],"branch\_id":null,"direction":"posterior","executable\_argument\_node\_ids":["LM:S:ST9:note:2:186-226"],"op":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:note:2:186-226"},"semantic\_operand\_id":"LM:S:ST9:note:2:186-226"]},"relation\_type":"relative-to"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:186-226"],"source\_statement\_id":"S:ST9:note:2"},"executable\_argument\_node\_ids":["LM:S:ST9:note:2:186-226"],"executor":"directional\_relation","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:note:2:186-226"},"semantic\_operand\_id":"LM:S:ST9:note:2:186-226"]},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:186-226"],"semantic\_fields":{"branch\_id":null,"direction":"posterior","op":"directional\_relation","relation\_type":"relative-to"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:186-226"],"source\_statement\_id":"S:ST9:note:2"},"semantic\_fields\_hash":"e79d950d98f6ddc102d92062d544beebcd09639cf4b54c6888e76b7f3c7b8"},"status":"RESOLVED"},"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"op":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:note:2:44-48"},"semantic\_operand\_id":"LM:S:ST9:note:2:44-48"]},"relation\_type":"same-level"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"source\_statement\_id":"S:ST9:note:2"},"executable\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"executor":"same\_level\_plane","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:note:2:44-48"},"semantic\_operand\_id":"LM:S:ST9:note:2:44-48"]},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"semantic\_fields":{"branch\_id":null,"op":"same\_level\_plane","relation\_type":"same-level"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"source\_statement\_id":"S:ST9:note:2"},"semantic\_fields\_hash":"05f75cd806b6e4801296c87e519e93e454c0c0cdc8c65cd6b235b958f41595a5"},"status":"RESOLVED"},"constraint":{"argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"branch\_id":null,"executable\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"op":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:note:2:44-48"},"semantic\_operand\_id":"LM:S:ST9:note:2:44-48"]},"relation\_type":"reference-acupoint"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"source\_statement\_id":"S:ST9:note:2"},"executable\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"executor":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM:S:ST9:note:2:44-48"},"semantic\_operand\_id":"LM:S:ST9:note:2:44-48"]},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"semantic\_fields":{"branch\_id":null,"op":"reference\_dependency","relation\_type":"reference-acupoint"},"semantic\_argument\_node\_ids":["LM:S:ST9:note:2:44-48"],"source\_statement\_id":"S:ST9:note:2"},"semantic\_fields\_hash":"aa27ce51039c4c30749b294c12b3e3374f453ff269d55a273d822456ff52fbc6"},"status":"RESOLVED"},"constraint":{"argument\_node\_ids":["LM21:S:ST9:note:2:50-54:6775aaafef"],"branch\_id":null,"executable\_argument\_node\_ids":["LM21:S:ST9:note:2:50-54:6775aaafef"],"op":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM21:S:ST9:note:2:50-54:6775aaafef"},"semantic\_operand\_id":"LM21:S:ST9:note:2:50-54:6775aaafef"]},"relation\_type":"reference-acupoint"},"semantic\_argument\_node\_ids":["LM21:S:ST9:note:2:50-54:6775aaafef"],"source\_statement\_id":"S:ST9:note:2"},"executable\_argument\_node\_ids":["LM21:S:ST9:note:2:50-54:6775aaafef"],"executor":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM21:S:ST9:note:2:50-54:6775aaafef"},"semantic\_operand\_id":"LM21:S:ST9:note:2:50-54:6775aaafef"]},"semantic\_argument\_node\_ids":["LM21:S:ST9:note:2:50-54:6775aaafef"],"semantic\_fields":{"branch\_id":null,"op":"reference\_dependency","relation\_type":"reference-acupoint"},"semantic\_argument\_node\_ids":["LM21:S:ST9:note:2:50-54:6775aaafef"],"source\_statement\_id":"S:ST9:note:2"},"semantic\_fields\_hash":"5d66dd3c9ac274d662e5599beb94f1d480f0a2c8a2087975cb690cfe0a33ab0c"},"status":"RESOLVED"},"constraint":{"argument\_node\_ids":["LM21:S:ST9:note:2:168-172:6775aaafef"],"branch\_id":null,"executable\_argument\_node\_ids":["LM21:S:ST9:note:2:168-172:6775aaafef"],"op":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM21:S:ST9:note:2:168-172:6775aaafef"},"semantic\_operand\_id":"LM21:S:ST9:note:2:168-172:6775aaafef"]},"relation\_type":"reference-acupoint"},"semantic\_argument\_node\_ids":["LM21:S:ST9:note:2:168-172:6775aaafef"],"source\_statement\_id":"S:ST9:note:2"},"semantic\_fields\_hash":"6dd689ff346688754baabc53626985eeeb2a0d81064fae5d42e3caa34f170894"},"status":"RESOLVED"},"constraint":{"argument\_node\_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"branch\_id":null,"executable\_argument\_node\_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"op":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM21:S:ST9:note:2:222-226:34574dfae8"},"semantic\_operand\_id":"LM21:S:ST9:note:2:222-226:34574dfae8"]},"relation\_type":"reference-acupoint"},"semantic\_argument\_node\_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"source\_statement\_id":"S:ST9:note:2"},"executable\_argument\_node\_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"executor":"reference\_dependency","operand\_binding\_trace":[{"binding":"identity","executable\_operand\_id":"LM21:S:ST9:note:2:222-226:34574dfae8"},"semantic\_operand\_id":"LM21:S:ST9:note:2:222-226:34574dfae8"]},"semantic\_argument\_node\_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"semantic\_fields":{"branch\_id":null,"op":"reference\_dependency","relation\_type":"reference-acupoint"},"semantic\_argument\_node\_ids":["LM21:S:ST9:note:2:222-226:34574dfae8"],"source\_statement\_id":"S:ST9:note:2"},"semantic\_fields\_hash":"9c70b37fa696f6c67ce297e0ce061a359e920e6442c815d6b0d1e98a086f20b2"},"status":"RESOLVED"},"status":"UNRESOLVED"}  
Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature
- [ ] distinct subfeatures
- [ ] relation operator
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# TE20: UNRESOLVED -> UNRESOLVED

Defect family: condition\_semantics\_loss

## WHO source

location | S:TE20:location | page 176 | On the head, just superior to the auricular apex.

note | S:TE20:note:1 | page 176 | Note: With the auricle folded forward and pressed against the head, the point is located where the auricular apex touches the head.

## Semantic decomposition

LM:S:TE20:location:29-48 | source= the auricular apex | class=soft\_tissue\_feature | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={ "concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:TE20:location:7-11 | source=head | class=region.body\_region | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={ "concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA7154","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

## Composite binding view

### Relation execution

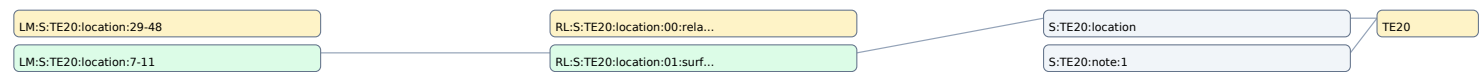
RL:S:TE20:location:00:relative-to | op=directional\_relation | semantic=[ 'LM:S:TE20:location:29-48' ] | executable=[] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:TE20:location:01:surface-landmark | op=surface\_membership | semantic=[ 'LM:S:TE20:location:7-11' ] | executable=[ 'LM:S:TE20:location:7-11' ] | dir=None | status=RESOLVED | reason=None

### Conditions

{ "condition\_id":"CD:S:TE20:note:1:00","condition\_type":"body\_position","default\_pose\_compatible":false,"reason":"condition requires pose incompatible with default BodyParts3D pose","repaired\_status":"CONDITIONAL","source\_raw":"With the auricle folded","source\_statement\_id":"S:TE20:note:1" }

### Dependency graph



### Dependency graph edge list

LM:S:TE20:location:7-11 -> RL:S:TE20:location:01:surface-landmark [executable\_to\_relation]

RL:S:TE20:location:01:surface-landmark -> S:TE20:location [relation\_to\_statement]

S:TE20:location -> TE20 [statement\_to\_point]

S:TE20:note:1 -> TE20 [statement\_to\_point]

### BEFORE / AFTER diff

BEFORE: { "conditions": [ { "branch\_id": "branch\_1", "condition\_type": "body\_position", "executor": "condition\_contract", "reason": null, "status": "RESOLVED" }, { "argument\_node\_ids": [ "LM:S:TE20:location:29-48" ], "executor": "directional\_relation", "reason": "argument landmark not uniquely executable", "status": "UNRESOLVED" }, { "constraint": { "argument\_node\_ids": [ "LM:S:TE20:location:7-11" ], "branch\_id": null, "op": "surface\_membership", "source\_statement\_id": "S:TE20:location" }, "executor": "surface\_membership", "status": "RESOLVED" }, { "status": "UNRESOLVED" } ] }

AFTER: { "conditions": [ { "branch\_id": "branch\_1", "condition\_type": "body\_position", "default\_pose\_compatible": false, "executor": "condition\_contract", "reason": "condition requires pose incompatible with default BodyParts3D pose", "source\_span": { "char\_end": 29, "char\_start": 6, "condition\_type": "body\_position", "role": null, "source\_raw": "With the auricle folded" }, "source\_statement\_id": "S:TE20:note:1", "status": "CONDITIONAL" }, { "argument\_node\_ids": [ "LM:S:TE20:location:29-48" ], "executor": "directional\_relation", "operand\_binding\_trace": [ { "binding": "missing\_executable\_subfeature", "executable\_operand\_ids": [ ], "semantic\_operand\_id": "LM:S:TE20:location:29-48" }, { "reason": "semantic composite has multiple child subfeatures and no unique executable operand", "semantic\_argument\_node\_ids": [ "LM:S:TE20:location:29-48" ], "semantic\_fields": { "branch\_id": null, "direction": "superior", "op": "directional\_relation", "relation\_type": "relative-to", "semantic\_argument\_node\_ids": [ "LM:S:TE20:location:29-48" ], "source\_statement\_id": "S:TE20:location" }, "semantic\_fields\_hash": "78e8dc4bd902c8a77ce9b262ee187bac1edf61c27a1a4aabfa2dd52e6e687e65", "status": "UNRESOLVED" }, { "constraint": { "argument\_node\_ids": [ "LM:S:TE20:location:7-11" ], "branch\_id": null, "executable\_argument\_node\_ids": [ "LM:S:TE20:location:7-11" ], "op": "surface\_membership", "operand\_binding\_trace": [ { "binding": "identity", "executable\_operand\_id": "LM:S:TE20:location:7-11", "semantic\_operand\_id": "LM:S:TE20:location:7-11" }, { "relation\_type": "surface-landmark", "semantic\_argument\_node\_ids": [ "LM:S:TE20:location:7-11" ], "source\_statement\_id": "S:TE20:location" }, "executable\_argument\_node\_ids": [ "LM:S:TE20:location:7-11" ], "executor": "surface\_membership", "operand\_binding\_trace": [ { "binding": "identity", "executable\_operand\_id": "LM:S:TE20:location:7-11", "semantic\_operand\_id": "LM:S:TE20:location:7-11" }, { "relation\_type": "surface-landmark", "semantic\_argument\_node\_ids": [ "LM:S:TE20:location:7-11" ], "source\_statement\_id": "S:TE20:location" }, "semantic\_fields\_hash": "629ee7cddb9cfce055ec9cabdfe1142b2a1ba75a09eb748ef73054d9e5a1cddb4", "status": "RESOLVED" }, { "status": "UNRESOLVED" } ] } ] }

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature
- [ ] distinct subfeatures
- [ ] relation operator
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# HT7: UNRESOLVED -> UNRESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:HT7:location | page 94 | On the anteromedial aspect of the wrist, radial to the flexor carpi ulnaris tendon, on the palmar wrist crease.

note | S:HT7:note:1 | page 94 | Note: In the depression radial to the proxi-mal border of the pisiform bone, on the palmar wrist crease.

## Semantic decomposition

LM:S:HT7:location:50-82 | source= the flexor carpi ulnaris tendon | class=tendon | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:HT7:location:7-26 | source=anteromedial aspect | class=surface.aspect | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"6621b519a3e9551fb3372da3e363c6dd82762c2bc01e5e66f461ac31124c6054","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:anteromedial\_wrist","subfeature\_type":null} | reason=None

LM:S:HT7:location:91-110 | source=palmar wrist crease | class=crease.skin\_crease | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:palm ar\_wrist\_crease","subfeature\_type":null} | reason=None

LM:S:HT7:note:1:13-23 | source=depression | class=fossa\_or\_depression | parent=None | child=None | executor=explicit\_unresolved\_taxonomy | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=ambiguous\_generic

LM:S:HT7:note:1:33-75 | source= the proxi-mal border of the pisiform bone | class=boundary.border | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:HT7:note:1:38-54 | source=proxi-mal border | class=boundary.border | parent=LM:S:HT7:note:1:58-75 | child=proxi-mal border | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=parent entity is not uniquely resolved

LM:S:HT7:note:1:58-75 | source=the pisiform bone | class=bone.bone | parent=None | child=None | executor=atlas\_name\_resolution | status=MULTIPLE | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=multiple atlas-name candidates

LM:S:HT7:note:1:84-103 | source=palmar wrist crease | class=crease.skin\_crease | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:palm ar\_wrist\_crease","subfeature\_type":null} | reason=None

## Composite binding view

PARENT LM:S:HT7:note:1:58-75 the pisiform bone status=MULTIPLE identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:HT7:note:1:38-54 proxi-mal border status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

## Relation execution

RL:S:HT7:location:00:relative-to | op=directional\_relation | semantic=['LM:S:HT7:location:50-82'] | executable=[] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:HT7:location:01:surface-landmark | op=surface\_membership | semantic=['LM:S:HT7:location:7-26'] | executable=['LM:S:HT7:location:7-26'] | dir=None | status=RESOLVED | reason=None

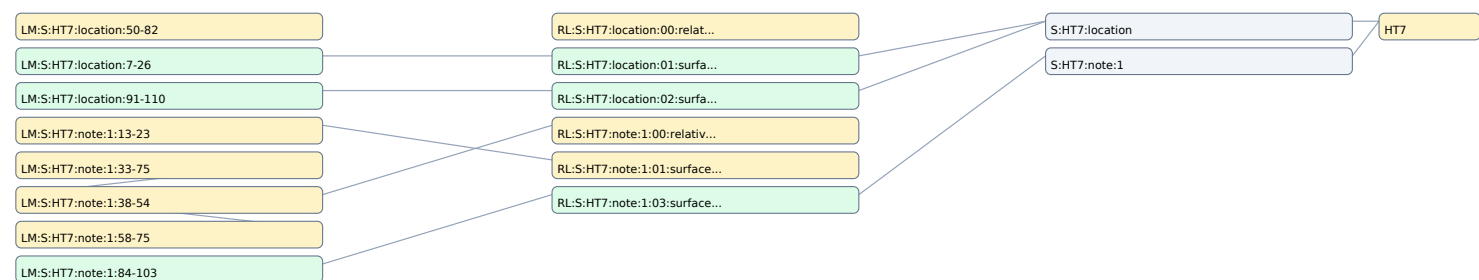
RL:S:HT7:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:HT7:location:91-110'] | executable=['LM:S:HT7:location:91-110'] | dir=None | status=RESOLVED | reason=None

RL:S:HT7:note:1:00:relative-to | op=directional\_relation | semantic=['LM:S:HT7:note:1:33-75'] | executable=['LM:S:HT7:note:1:38-54'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:HT7:note:1:01:surface-landmark | op=surface\_membership | semantic=['LM:S:HT7:note:1:13-23'] | executable=['LM:S:HT7:note:1:13-23'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:HT7:note:1:03:surface-landmark | op=surface\_membership | semantic=['LM:S:HT7:note:1:84-103'] | executable=['LM:S:HT7:note:1:84-103'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:HT7:location:7-26 -> RL:S:HT7:location:01:surface-landmark [executable\_to\_relation]  
 RL:S:HT7:location:01:surface-landmark -> S:HT7:location [relation\_to\_statement]  
 LM:S:HT7:location:91-110 -> RL:S:HT7:location:02:surface-landmark [executable\_to\_relation]  
 RL:S:HT7:location:02:surface-landmark -> S:HT7:location [relation\_to\_statement]  
 LM:S:HT7:note:1:33-75 -> LM:S:HT7:note:1:38-54 [semantic\_to\_executable]  
 LM:S:HT7:note:1:38-54 -> RL:S:HT7:note:1:00:relative-to [executable\_to\_relation]  
 LM:S:HT7:note:1:13-23 -> RL:S:HT7:note:1:01:surface-landmark [executable\_to\_relation]  
 LM:S:HT7:note:1:84-103 -> RL:S:HT7:note:1:03:surface-landmark [executable\_to\_relation]  
 RL:S:HT7:note:1:03:surface-landmark -> S:HT7:note:1 [relation\_to\_statement]  
 LM:S:HT7:note:1:58-75 -> LM:S:HT7:note:1:38-54 [parent\_to\_child\_subfeature]  
 S:HT7:location -> HT7 [statement\_to\_point]  
 S:HT7:note:1 -> HT7 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

[ ] WHO source phrase 0000 0000  
 [ ] semantic granularity 0000  
 [ ] child subfeature 00 00 execution 0000  
 [ ] distinct subfeatures 00 distinct executable identity 00 00  
 [ ] relation operator 0000 operand 0000  
 [ ] condition state 0000 0000  
 [ ] RESOLVED/UNRESOLVED 0000 0000  
 [ ] ACCEPT  
 [ ] REJECT

# LI4: RESOLVED -> UNRESOLVED

Defect family: parent\_only\_composite\_resolution, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, parent\_only\_composite\_resolution, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:LI4:location | page 44 | On the dorsum of the hand, radial to the midpoint of the second metacarpal bone.

## Semantic decomposition

LM:S:LI4:location:36-79 | source= the midpoint of the second metacarpal bone | class=bone.bone | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:LI4:location:53-79 | source=the second metacarpal bone | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA23900","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:LI4:location:7-25 | source=dorsum of the hand | class=surface.aspect | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"660965ac347a4b5e19c1f02c3d45a8be59be919076604c2a490ce1a7242bf5f4","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:dorsum\_hand","subfeature\_type":null} | reason=None

## Composite binding view

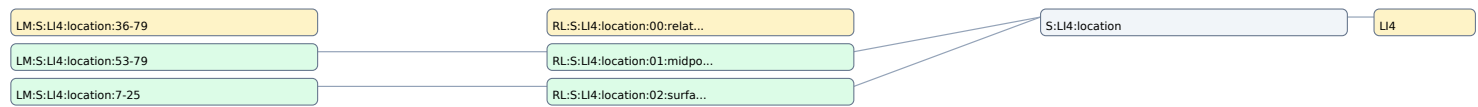
### Relation execution

RL:S:LI4:location:00:relative-to | op=directional\_relation | semantic=['LM:S:LI4:location:36-79'] | executable=[] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:LI4:location:01:midpoint-of-entity | op=entity\_midpoint | semantic=['LM:S:LI4:location:53-79'] | executable=['LM:S:LI4:location:53-79'] | dir=None | status=RESOLVED | reason=None

RL:S:LI4:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:LI4:location:7-25'] | executable=['LM:S:LI4:location:7-25'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:LI4:location:53-79 -> RL:S:LI4:location:01:midpoint-of-entity [executable\_to\_relation]

RL:S:LI4:location:01:midpoint-of-entity -> S:LI4:location [relation\_to\_statement]

LM:S:LI4:location:7-25 -> RL:S:LI4:location:02:surface-landmark [executable\_to\_relation]

RL:S:LI4:location:02:surface-landmark -> S:LI4:location [relation\_to\_statement]

S:LI4:location -> LI4 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature execution
- [ ] distinct subfeatures distinct executable identity
- [ ] relation operator operand
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# ST1: UNRESOLVED -> UNRESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:ST1:location | page 55 | On the face, between the eyeball and the infraorbital margin, directly inferior to the pupil.

## Semantic decomposition

LM:S:ST1:location:25-32 | source=eyeball | class=soft\_tissue\_feature | parent=None | child=None | executor=atlas\_name\_resolution | status=MULTIPLE | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=multiple atlas-name candidates

LM:S:ST1:location:37-60 | source=the infraorbital margin | class=boundary.border | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST1:location:7-11 | source=face | class=region.body\_region | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA24728","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST1:location:82-92 | source= the pupil | class=soft\_tissue\_feature | parent=None | child=None | executor=no\_family\_executor | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=unsupported or unresolved landmark: geometry\_or\_surface/soft\_tissue\_feature

## Composite binding view

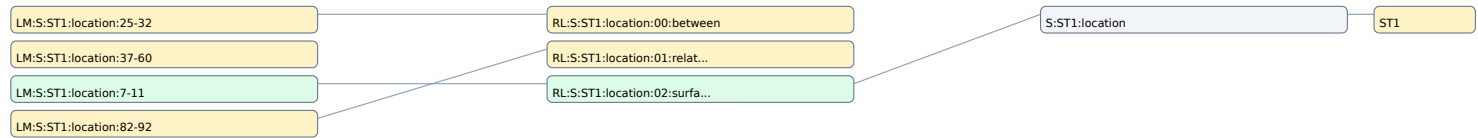
### Relation execution

RL:S:ST1:location:00:between | op=between\_constraint | semantic=['LM:S:ST1:location:25-32', 'LM:S:ST1:location:37-60'] | executable=['LM:S:ST1:location:25-32'] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:ST1:location:01:relative-to | op=directional\_relation | semantic=['LM:S:ST1:location:82-92'] | executable=['LM:S:ST1:location:82-92'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST1:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:ST1:location:7-11'] | executable=['LM:S:ST1:location:7-11'] | dir=None | status=RESOLVED | reason=None

### Dependency graph



### Dependency graph edge list

- LM:S:ST1:location:25-32 -> RL:S:ST1:location:00:between [executable\_to\_relation]
- LM:S:ST1:location:82-92 -> RL:S:ST1:location:01:relative-to [executable\_to\_relation]
- LM:S:ST1:location:7-11 -> RL:S:ST1:location:02:surface-landmark [executable\_to\_relation]
- RL:S:ST1:location:02:surface-landmark -> S:ST1:location [relation\_to\_statement]
- S:ST1:location -> ST1 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature execution
- [ ] distinct subfeatures distinct executable identity
- [ ] relation operator operand
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# GB14: UNRESOLVED -> UNRESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:GB14:location | page 187 | On the head, 1 B-cun superior to the eye-brow, directly superior to the centre of the pupil.

## Semantic decomposition

LM:S:GB14:location:32-45 | source= the eye-brow | class=soft\_tissue\_feature | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA54237","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:GB14:location:67-91 | source= the centre of the pupil | class=soft\_tissue\_feature | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:GB14:location:7-11 | source=head | class=region.body\_region | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA7154","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:GB14:location:72-78 | source=centre | class=None | parent=LM:S:GB14:location:82-91 | child=centre | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=parent entity is not uniquely resolved

LM:S:GB14:location:82-91 | source=the pupil | class=soft\_tissue\_feature | parent=None | child=None | executor=no\_family\_executor | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=unsupported or unresolved landmark: geometry\_or\_surface/soft\_tissue\_feature

## Composite binding view

PARENT LM:S:GB14:location:82-91 the pupil status=UNRESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:GB14:location:72-78 centre status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

## Relation execution

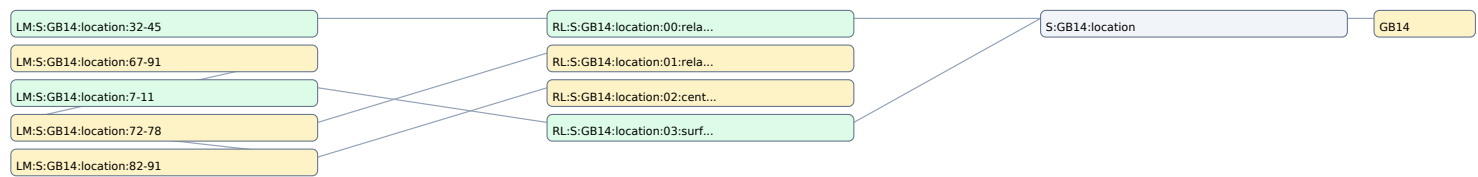
RL:S:GB14:location:00:relative-to | op=directional\_relation | semantic=['LM:S:GB14:location:32-45'] | executable=['LM:S:GB14:location:32-45'] | dir=superior | status=RESOLVED | reason=None

RL:S:GB14:location:01:relative-to | op=directional\_relation | semantic=['LM:S:GB14:location:67-91'] | executable=['LM:S:GB14:location:72-78'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:GB14:location:02:center-of | op=entity\_center | semantic=['LM:S:GB14:location:82-91'] | executable=['LM:S:GB14:location:82-91'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:GB14:location:03:surface-landmark | op=surface\_membership | semantic=['LM:S:GB14:location:7-11'] | executable=['LM:S:GB14:location:7-11'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

- LM:S:GB14:location:32-45 -> RL:S:GB14:location:00:relative-to [executable\_to\_relation]
- RL:S:GB14:location:00:relative-to -> S:GB14:location [relation\_to\_statement]
- LM:S:GB14:location:67-91 -> LM:S:GB14:location:72-78 [semantic\_to\_executable]
- LM:S:GB14:location:72-78 -> RL:S:GB14:location:01:relative-to [executable\_to\_relation]
- LM:S:GB14:location:82-91 -> RL:S:GB14:location:02:center-of [executable\_to\_relation]
- LM:S:GB14:location:7-11 -> RL:S:GB14:location:03:surface-landmark [executable\_to\_relation]
- RL:S:GB14:location:03:surface-landmark -> S:GB14:location [relation\_to\_statement]
- LM:S:GB14:location:82-91 -> LM:S:GB14:location:72-78 [parent\_to\_child\_subfeature]
- S:GB14:location -> GB14 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase 何者 何
- [ ] semantic granularity 意味
- [ ] child subfeature 子特徴 execution 実行
- [ ] distinct subfeatures 異なる実行可能 identity 同一
- [ ] relation operator 関係演算器 operand 演算子
- [ ] condition state 条件状態
- [ ] RESOLVED/UNRESOLVED 解決済み/未解決
- [ ] ACCEPT 受入れ
- [ ] REJECT 拒絶

# GB23: RESOLVED -> RESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:GB23:location | page 192 | In the lateral thoracic region, in the fourth intercostal space, 1 B-cun anterior to the midaxillary line.

## Semantic decomposition

LM:S:GB23:location:39-63 | source=fourth intercostal space | class=space.anatomical\_space | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"c8e8691e4f765638eb499518792d13bea202f0a5aa3a1f90a050c51fb2e2ed36","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:fourth\_intercostal\_space","subfeature\_type":null} | reason=None

LM:S:GB23:location:7-30 | source=lateral thoracic region | class=region.body\_region | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"8cf739f6e831150ec0c56595886dd8d46bc412a09ff8ff132ad1ae55e653d9da","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:latera\_l\_thorax","subfeature\_type":null} | reason=None

LM:S:GB23:location:84-105 | source= the midaxillary line | class=line.anatomical\_line | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"643f43e0388878b8eb9f6175a1c685aeb1b7b80df0274d8adc5b0b25cfda1420","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:mid axillary\_line","subfeature\_type":null} | reason=None

## Composite binding view

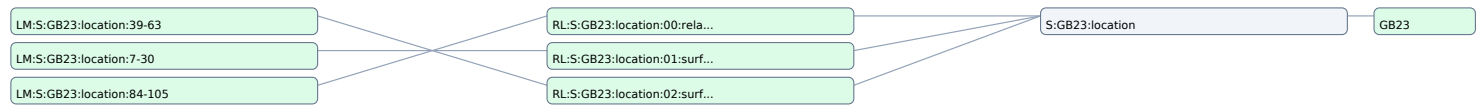
## Relation execution

RL:S:GB23:location:00:relative-to | op=directional\_relation | semantic=['LM:S:GB23:location:84-105'] | executable=['LM:S:GB23:location:84-105'] | dir=anterior | status=RESOLVED | reason=None

RL:S:GB23:location:01:surface-landmark | op=surface\_membership | semantic=['LM:S:GB23:location:7-30'] | executable=['LM:S:GB23:location:7-30'] | dir=None | status=RESOLVED | reason=None

RL:S:GB23:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:GB23:location:39-63'] | executable=['LM:S:GB23:location:39-63'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:GB23:location:84-105 -> RL:S:GB23:location:00:relative-to [executable\_to\_relation]  
RL:S:GB23:location:00:relative-to -> S:GB23:location [relation\_to\_statement]  
LM:S:GB23:location:7-30 -> RL:S:GB23:location:01:surface-landmark [executable\_to\_relation]  
RL:S:GB23:location:01:surface-landmark -> S:GB23:location [relation\_to\_statement]  
LM:S:GB23:location:39-63 -> RL:S:GB23:location:02:surface-landmark [executable\_to\_relation]  
RL:S:GB23:location:02:surface-landmark -> S:GB23:location [relation\_to\_statement]  
S:GB23:location -> GB23 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature execution
- [ ] distinct subfeatures distinct executable identity
- [ ] relation operator operand
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

## LU6: UNRESOLVED -> UNRESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss,  
relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss,  
relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure,  
relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss,  
relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch,  
operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss,  
relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss,  
relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch,  
operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss,  
relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete,  
relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure,  
relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss,  
relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete,  
relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch,  
operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss,  
relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete,  
relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss,  
relation semantic hash mismatch, operand trace incomplete

WHO source

location | S:LU6:location | page 37 | On the anterolateral aspect of the forearm, on the line connecting LU5 with LU9, 7 B-cun superior to the palmar wrist crease.

note | S:LU6:note:1 | page 37 | Note: LU6 is 5 B-cun inferior to LU5, 1 B-cun superior to the midpoint of the line connecting LU5 with LU9.

## Semantic decomposition

LM:S:LU6:location:100-124 | source= the palmar wrist crease | class=crease.skin\_crease | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:palmar\_wrist\_crease","subfeature\_type":null} | reason=None

```
LM:S:LU6:location:51-66 | source=line connecting | class=line.reference_line | parent=None | child=None | executor=no_family_executor |
status=UNRESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":null,"kind":null,"parent_landma
rk_id":null,"part_id":null,"reference_point_id":null,"registry_id":null,"subfeature_type":null} | reason=unsupported or unresolved landmark:
geometry_or_surface/line.reference_line
```

```
LM:S:LU6:location:67-70 | source=LU5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED |
nullity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":
null,"part_id":null,"reference_point_id":"LU5","registry_id":null,"subfeature_type":null} | reason=None
```

```
LM:S:LU6:location:7-27 | source=anterolateral aspect | class=surface.aspect | parent=None | child=None | executor=approved_surface_registry |
status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":"8ce43d41eaba7ffc905ab399e62
1546e96abede4af22e20655f5814a9f8b665c","kind":"surface_registry","parent_landmark_id":null,"part_id":null,"reference_point_id":null,"registry_id":"SR:anter
olateral forearm"},"subfeature_type":null | reason=None
```

```
LM:S:LU6:location:76-79 | source=LU9 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED |
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":
null,"part_id":null,"reference_point_id":"LU9","registry_id":null,"subfeature_type":null} | reason=None
```

```
LM:S:LU6:note:1:103-106 | source=LU9 | class=acupoint | reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED |
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":
null,"part_id":null,"reference_point_id":"LU9","registry_id":null,"subfeature_type":null} | reason=None
```

```
LM:S:LU6:note:1:32-36 | source= LU5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED |
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":
null,"part_id":null,"reference_point_id":"LU5","registry_id":null,"subfeature_type":null} | reason=None
```

LM:S:LU6:note:1:57-106 | source= the midpoint of the line connecting LU5 with LU9 | class=line.reference\_line | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

```
LM:S:LU6:note:1:94-97 | source=LU5 | class=acupoint | reference | parent=None | child=None | executor=reference_acupoint | status=RESOLVED |
identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","parent_landmark_id":
null,"part_id":null,"reference_point_id":"LU5","registry_id":null,"subfeature_type":null} | reason=None
```

```
LM21:S:LU6:location:67-70:002c0a95e5 source=LU5 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint |
status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint","
parent_landmark_id":null,"part_id":null,"reference_point_id":"LU5","registry_id":null,"subfeature_type":null} | reason=None
```

```
LM21:S:LU6:location:76-79:30c4b43ed7 | source=LU9 | class=acupoint_reference | parent=None | child=None | executor=reference_acupoint |
status=RESOLVED | identity={"concept_id":null,"constructed_id":null,"construction_rule":null,"fma_id":null,"geometry_hash":null,"kind":"reference_acupoint",
parent_landmark_id":null,"part_id":null,"reference_point_id":"LU9","registry_id":null,"subfeature_type":null} | reason=None
```

## Composite binding view

## Relation execution

RL:S:LU6:location:00:on-line | op=line\_membership | semantic=['LM:S:LU6:location:67-70', 'LM:S:LU6:location:76-79'] | executable=['LM:S:LU6:location:67-70', 'LM:S:LU6:location:76-79'] | dir=None | status=RESOLVED | reason=None

RL:S:LU6:location:01:relative-to | op=directional\_relation | semantic=['LM:S:LU6:location:100-124'] | executable=['LM:S:LU6:location:100-124'] | dir=superior | status=RESOLVED | reason=None

RL:S:LU6:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:LU6:location:7-27'] | executable=['LM:S:LU6:location:7-27'] | dir=None | status=RESOLVED | reason=None

RL:S:LU6:location:03:surface-landmark | op=surface\_membership | semantic=['LM:S:LU6:location:51-66'] | executable=['LM:S:LU6:location:51-66'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:LU6:note:1:00:on-line | op=line\_membership | semantic=['LM:S:LU6:note:1:94-97', 'LM:S:LU6:note:1:103-106'] | executable=['LM:S:LU6:note:1:94-97', 'LM:S:LU6:note:1:103-106'] | dir=None | status=RESOLVED | reason=None

RL:S:LU6:note:1:01:midpoint-between | op=midpoint\_between | semantic=['LM:S:LU6:note:1:94-97', 'LM:S:LU6:note:1:103-106'] | executable=['LM:S:LU6:note:1:94-97', 'LM:S:LU6:note:1:103-106'] | dir=None | status=RESOLVED | reason=None

RL:S:LU6:note:1:02:relative-to | op=directional\_relation | semantic=['LM:S:LU6:note:1:32-36'] | executable=['LM:S:LU6:note:1:32-36'] | dir=inferior | status=RESOLVED | reason=None

RL:S:LU6:note:1:03:relative-to | op=directional\_relation | semantic=['LM:S:LU6:note:1:57-106'] | executable=[] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 | op=reference\_dependency | semantic=['LM21:S:LU6:location:67-70:002c0a95e5'] | executable=['LM21:S:LU6:location:67-70:002c0a95e5'] | dir=None | status=RESOLVED | reason=None

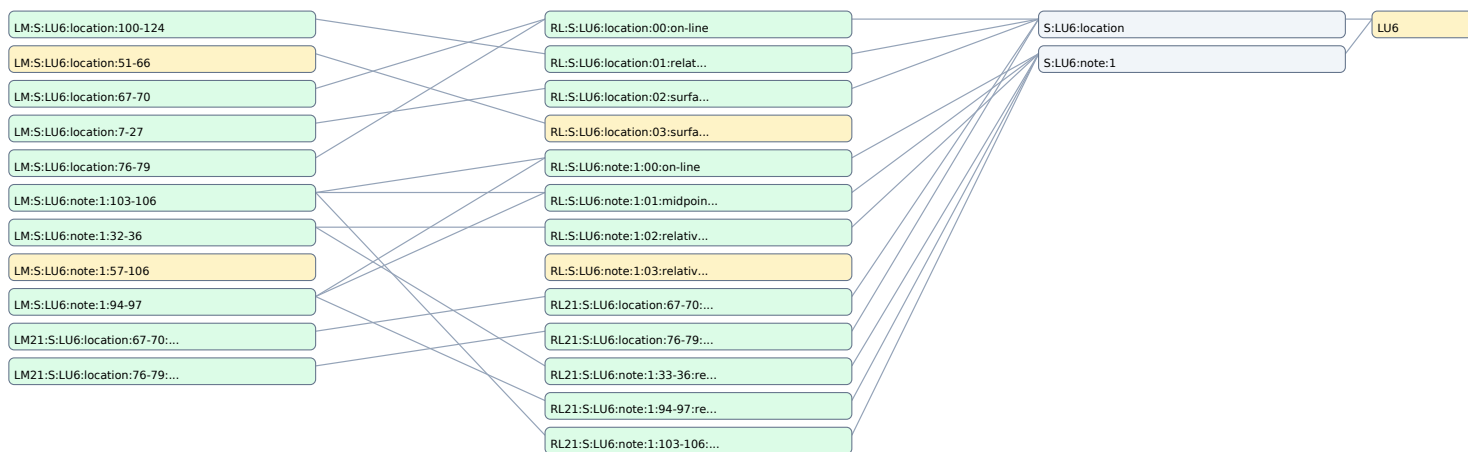
RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 | op=reference\_dependency | semantic=['LM21:S:LU6:location:76-79:30c4b43ed7'] | executable=['LM21:S:LU6:location:76-79:30c4b43ed7'] | dir=None | status=RESOLVED | reason=None

RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 | op=reference\_dependency | semantic=['LM:S:LU6:note:1:32-36'] | executable=['LM:S:LU6:note:1:32-36'] | dir=None | status=RESOLVED | reason=None

RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b | op=reference\_dependency | semantic=['LM:S:LU6:note:1:94-97'] | executable=['LM:S:LU6:note:1:94-97'] | dir=None | status=RESOLVED | reason=None

RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 | op=reference\_dependency | semantic=['LM:S:LU6:note:1:103-106'] | executable=['LM:S:LU6:note:1:103-106'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:LU6:location:67-70 -> RL:S:LU6:location:00:on-line [executable\_to\_relation]

LM:S:LU6:location:76-79 -> RL:S:LU6:location:00:on-line [executable\_to\_relation]

RL:S:LU6:location:00:on-line -> S:LU6:location [relation\_to\_statement]

LM:S:LU6:location:100-124 -> RL:S:LU6:location:01:relative-to [executable\_to\_relation]

RL:S:LU6:location:01:relative-to -> S:LU6:location [relation\_to\_statement]

LM:S:LU6:location:7-27 -> RL:S:LU6:location:02:surface-landmark [executable\_to\_relation]

RL:S:LU6:location:02:surface-landmark -> S:LU6:location [relation\_to\_statement]

LM:S:LU6:location:51-66 -> RL:S:LU6:location:03:surface-landmark [executable\_to\_relation]

LM:S:LU6:note:1:94-97 -> RL:S:LU6:note:1:00:on-line [executable\_to\_relation]

LM:S:LU6:note:1:103-106 -> RL:S:LU6:note:1:00:on-line [executable\_to\_relation]

RL:S:LU6:note:1:00:on-line -> S:LU6:note:1 [relation\_to\_statement]

LM:S:LU6:note:1:94-97 -> RL:S:LU6:note:1:01:midpoint-between [executable\_to\_relation]

LM:S:LU6:note:1:103-106 -> RL:S:LU6:note:1:01:midpoint-between [executable\_to\_relation]

RL:S:LU6:note:1:01:midpoint-between -> S:LU6:note:1 [relation\_to\_statement]

LM:S:LU6:note:1:32-36 -> RL:S:LU6:note:1:02:relative-to [executable\_to\_relation]

RL:S:LU6:note:1:02:relative-to -> S:LU6:note:1 [relation\_to\_statement]

LM21:S:LU6:location:67-70:002c0a95e5 -> RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 [executable\_to\_relation]

RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 -> S:LU6:location [relation\_to\_statement]  
LM21:S:LU6:location:76-79:30c4b43ed7 -> RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 [executable\_to\_relation]  
RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 -> S:LU6:location [relation\_to\_statement]  
LM:S:LU6:note:1:32-36 -> RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 [executable\_to\_relation]  
RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 -> S:LU6:note:1 [relation\_to\_statement]  
LM:S:LU6:note:1:94-97 -> RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b [executable\_to\_relation]  
RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b -> S:LU6:note:1 [relation\_to\_statement]  
LM:S:LU6:note:1:103-106 -> RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 [executable\_to\_relation]  
RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 -> S:LU6:note:1 [relation\_to\_statement]  
S:LU6:location -> LU6 [statement\_to\_point]  
S:LU6:note:1 -> LU6 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

[ ] WHO source phrase 0000 0000  
[ ] semantic granularity 0000  
[ ] child subfeature 000 execution 0000  
[ ] distinct subfeatures 0 distinct executable identity 000  
[ ] relation operator 0000 operand 0000  
[ ] condition state 0000 0000  
[ ] RESOLVED/UNRESOLVED 0000 0000  
[ ] ACCEPT  
[ ] REJECT

# LI7: UNRESOLVED -> UNRESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:LI7:location | page 46 | On the posterolateral aspect of the forearm, on the line connecting LI5 with LI11, 5 B-cun superior to the dorsal wrist crease.

## Semantic decomposition

LM:S:LI7:location:102-126 | source= the dorsal wrist crease | class=crease.skin\_crease | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"0ff534739841ae12b463e9ea5102d7801e5de256871212f7ddeb84a0b0265288","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:dorsal\_wrist\_crease","subfeature\_type":null} | reason=None

LM:S:LI7:location:52-67 | source=line connecting | class=line.reference\_line | parent=None | child=None | executor=no\_family\_executor | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=unsupported or unresolved landmark: geometry\_or\_surface/line.reference\_line

LM:S:LI7:location:68-71 | source=LI5 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"LI5","registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:LI7:location:7-28 | source=posterolateral aspect | class=surface.aspect | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"4e542fb31cd71e7a148b3607b7424ea18ab74b0b8ddb0210ce3c5abccc2c54d9","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:posterolateral\_forearm","subfeature\_type":null} | reason=None

LM:S:LI7:location:77-81 | source=LI11 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"LI11","registry\_id":null,"subfeature\_type":null} | reason=None

LM21:S:LI7:location:68-71:1bff496df7 | source=LI5 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"LI5","registry\_id":null,"subfeature\_type":null} | reason=None

LM21:S:LI7:location:77-81:245ea01482 | source=LI11 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"LI11","registry\_id":null,"subfeature\_type":null} | reason=None

## Composite binding view

## Relation execution

RL:S:LI7:location:00:on-line | op=line\_membership | semantic=['LM:S:LI7:location:68-71', 'LM:S:LI7:location:77-81'] | executable=['LM:S:LI7:location:68-71', 'LM:S:LI7:location:77-81'] | dir=None | status=RESOLVED | reason=None

RL:S:LI7:location:01:relative-to | op=directional\_relation | semantic=['LM:S:LI7:location:102-126'] | executable=['LM:S:LI7:location:102-126'] | dir=superior | status=RESOLVED | reason=None

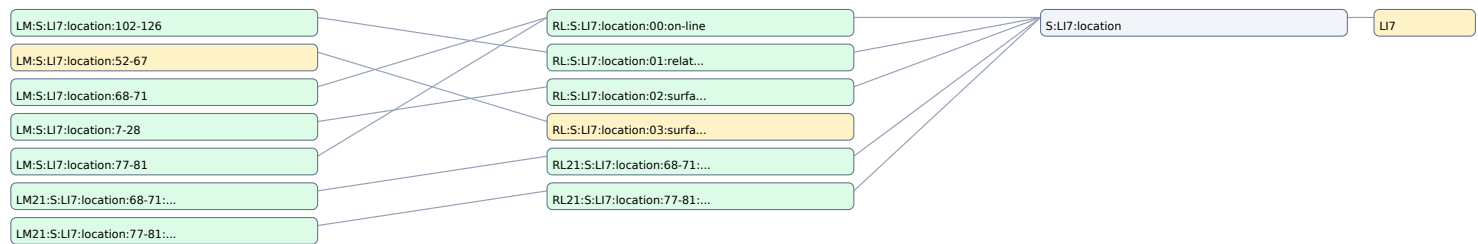
RL:S:LI7:location:02:surface-landmark | op=surface\_membership | semantic=['LM:S:LI7:location:7-28'] | executable=['LM:S:LI7:location:7-28'] | dir=None | status=RESOLVED | reason=None

RL:S:LI7:location:03:surface-landmark | op=surface\_membership | semantic=['LM:S:LI7:location:52-67'] | executable=['LM:S:LI7:location:52-67'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd | op=reference\_dependency | semantic=['LM21:S:LI7:location:68-71:1bff496df7'] | executable=['LM21:S:LI7:location:68-71:1bff496df7'] | dir=None | status=RESOLVED | reason=None

RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c | op=reference\_dependency | semantic=['LM21:S:LI7:location:77-81:245ea01482'] | executable=['LM21:S:LI7:location:77-81:245ea01482'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:LI7:location:68-71 -> RL:S:LI7:location:00:on-line [executable\_to\_relation]  
LM:S:LI7:location:77-81 -> RL:S:LI7:location:00:on-line [executable\_to\_relation]  
RL:S:LI7:location:00:on-line -> S:LI7:location [relation\_to\_statement]  
LM:S:LI7:location:102-126 -> RL:S:LI7:location:01:relative-to [executable\_to\_relation]  
RL:S:LI7:location:01:relative-to -> S:LI7:location [relation\_to\_statement]  
LM:S:LI7:location:7-28 -> RL:S:LI7:location:02:surface-landmark [executable\_to\_relation]  
RL:S:LI7:location:02:surface-landmark -> S:LI7:location [relation\_to\_statement]  
LM:S:LI7:location:52-67 -> RL:S:LI7:location:03:surface-landmark [executable\_to\_relation]  
LM21:S:LI7:location:68-71:1bff496df7 -> RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd [executable\_to\_relation]  
RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd -> S:LI7:location [relation\_to\_statement]  
LM21:S:LI7:location:77-81:245ea01482 -> RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c [executable\_to\_relation]  
RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c -> S:LI7:location [relation\_to\_statement]  
S:LI7:location -> LI7 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

[ ] WHO source phrase 0000 0000  
[ ] semantic granularity 0000  
[ ] child subfeature 0000 execution 0000  
[ ] distinct subfeatures 0000 distinct executable identity 0000  
[ ] relation operator 0000 operand 0000  
[ ] condition state 0000 0000  
[ ] RESOLVED/UNRESOLVED 0000 0000  
[ ] ACCEPT  
[ ] REJECT

# GB26: UNRESOLVED -> UNRESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:GB26:location | page 193 | On the lateral abdomen, inferior to the free extremity of the 11th rib, at the same level as the centre of umbilicus.

note | S:GB26:note:1 | page 193 | Note 1: GB26 can be located by first locating the tenth rib, then searching for the free extremity of the 11th rib that is situated immediately below the inferior border of the costal arch.

note | S:GB26:note:2 | page 193 | Note 2: GB26 is inferior to LR13, at the same level as CV8.

## Semantic decomposition

LM:S:GB26:location:107-116 | source=umbilicus | class=orifice\_or\_cavity | parent=None | child=None | executor=atlas\_name\_resolution | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no unique atlas name match

LM:S:GB26:location:35-70 | source= the free extremity of the 11th rib | class=structure.part | parent=None | child=None | executor=atlas\_name\_resolution | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no unique atlas name match

LM:S:GB26:location:40-54 | source=free extremity | class=structure.part | parent=LM:S:GB26:location:58-70 | child=free extremity | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=child subfeature type not recognized

LM:S:GB26:location:58-70 | source=the 11th rib | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA8499","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:GB26:location:7-22 | source=lateral abdomen | class=region.body\_region | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"912678a19ed1d94961027afd11c1d686ebc1dd7d67fe77a92a26935f99509145","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:lateral\_abdomen","subfeature\_type":null} | reason=None

LM:S:GB26:location:93-116 | source=the centre of umbilicus | class=orifice\_or\_cavity | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:GB26:location:97-103 | source=centre | class=None | parent=LM:S:GB26:location:107-116 | child=centre | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=parent entity is not uniquely resolved

LM:S:GB26:note:1:103-189 | source=the 11th rib that is situated immediately below the inferior border of the costal arch | class=boundary.border | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:GB26:note:1:85-99 | source=free extremity | class=structure.part | parent=LM:S:GB26:note:1:103-189 | child=free extremity | executor=composite\_subfeature\_binding | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=child subfeature type not recognized

LM:S:GB26:note:2:27-32 | source= LR13 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"LR13","registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:GB26:note:2:55-58 | source=CV8 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"CV8","registry\_id":null,"subfeature\_type":null} | reason=None

## Composite binding view

PARENT LM:S:GB26:location:58-70 the 11th rib status=RESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA8499","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:GB26:location:40-54 free extremity status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

PARENT LM:S:GB26:location:107-116 umbilicus status=UNRESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:GB26:location:97-103 centre status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

PARENT LM:S:GB26:note:1:103-189 the 11th rib that is situated immediately below the inferior border of the costal arch status=UNRESOLVED identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

-> CHILD LM:S:GB26:note:1:85-99 free extremity status=UNRESOLVED executable={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null}

## Relation execution

RL:S:GB26:location:00:relative-to | op=directional\_relation | semantic=['LM:S:GB26:location:35-70'] | executable=['LM:S:GB26:location:35-70'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:GB26:location:01:same-level | op=same\_level\_plane | semantic=['LM:S:GB26:location:93-116'] | executable=['LM:S:GB26:location:97-103'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:GB26:location:02:center-of | op=entity\_center | semantic=['LM:S:GB26:location:107-116'] | executable=['LM:S:GB26:location:107-116'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:GB26:location:03:surface-landmark | op=surface\_membership | semantic=['LM:S:GB26:location:7-22'] | executable=['LM:S:GB26:location:7-22'] | dir=None | status=RESOLVED | reason=None

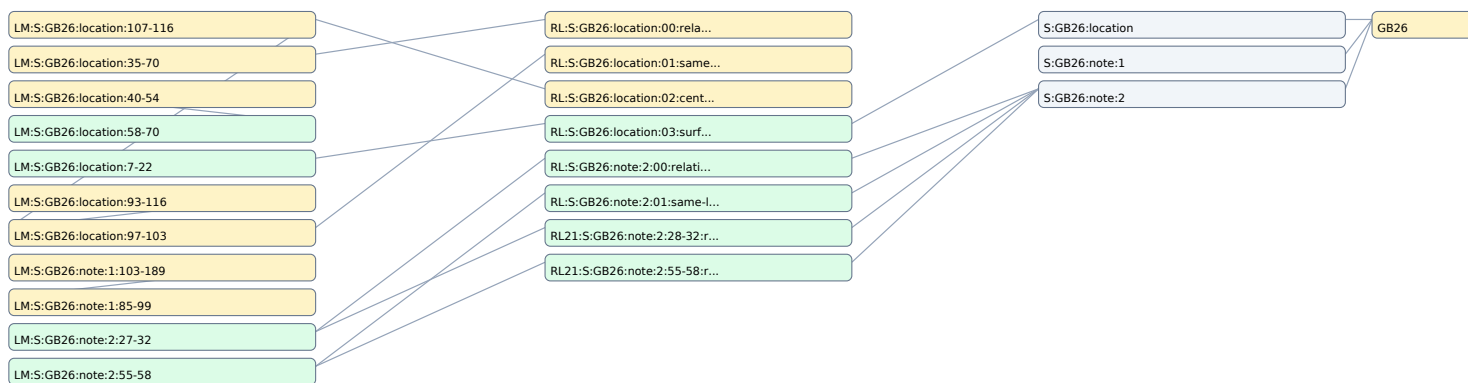
RL:S:GB26:note:2:00:relative-to | op=directional\_relation | semantic=['LM:S:GB26:note:2:27-32'] | executable=['LM:S:GB26:note:2:27-32'] | dir=inferior | status=RESOLVED | reason=None

RL:S:GB26:note:2:01:same-level | op=same\_level\_plane | semantic=['LM:S:GB26:note:2:55-58'] | executable=['LM:S:GB26:note:2:55-58'] | dir=None | status=RESOLVED | reason=None

RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 | op=reference\_dependency | semantic=['LM:S:GB26:note:2:27-32'] | executable=['LM:S:GB26:note:2:27-32'] | dir=None | status=RESOLVED | reason=None

RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d | op=reference\_dependency | semantic=['LM:S:GB26:note:2:55-58'] | executable=['LM:S:GB26:note:2:55-58'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:GB26:location:35-70 -> RL:S:GB26:location:00:relative-to [executable\_to\_relation]

LM:S:GB26:location:93-116 -> LM:S:GB26:location:97-103 [semantic\_to\_executable]

LM:S:GB26:location:97-103 -> RL:S:GB26:location:01:same-level [executable\_to\_relation]

LM:S:GB26:location:107-116 -> RL:S:GB26:location:02:center-of [executable\_to\_relation]

LM:S:GB26:location:7-22 -> RL:S:GB26:location:03:surface-landmark [executable\_to\_relation]

RL:S:GB26:location:03:surface-landmark -> S:GB26:location [relation\_to\_statement]

LM:S:GB26:note:2:27-32 -> RL:S:GB26:note:2:00:relative-to [executable\_to\_relation]

RL:S:GB26:note:2:00:relative-to -> S:GB26:note:2 [relation\_to\_statement]

LM:S:GB26:note:2:55-58 -> RL:S:GB26:note:2:01:same-level [executable\_to\_relation]

RL:S:GB26:note:2:01:same-level -> S:GB26:note:2 [relation\_to\_statement]

LM:S:GB26:note:2:27-32 -> RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 [executable\_to\_relation]

RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 -> S:GB26:note:2 [relation\_to\_statement]

LM:S:GB26:note:2:55-58 -> RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d [executable\_to\_relation]

RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d -> S:GB26:note:2 [relation\_to\_statement]

LM:S:GB26:location:58-70 -> LM:S:GB26:location:40-54 [parent\_to\_child\_subfeature]

LM:S:GB26:location:107-116 -> LM:S:GB26:location:97-103 [parent\_to\_child\_subfeature]

LM:S:GB26:note:1:103-189 -> LM:S:GB26:note:1:85-99 [parent\_to\_child\_subfeature]

S:GB26:location -> GB26 [statement\_to\_point]

S:GB26:note:1 -> GB26 [statement\_to\_point]

S:GB26:note:2 -> GB26 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature execution
- [ ] distinct subfeatures distinct executable identity
- [ ] relation operator operand
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# ST2: UNRESOLVED -> UNRESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:ST2:location | page 55 | On the face, in the infraorbital foramen.

## Semantic decomposition

LM:S:ST2:location:20-40 | source=infraorbital foramen | class=bone.opening | parent=None | child=None | executor=atlas\_name\_resolution | status=UNRESOLVED | identity={ "concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=no unique atlas name match

LM:S:ST2:location:7-11 | source=face | class=region.body\_region | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={ "concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA24728","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

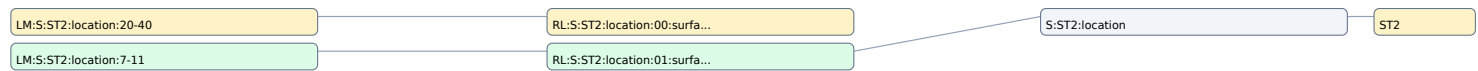
## Composite binding view

## Relation execution

RL:S:ST2:location:00:surface-landmark | op=surface\_membership | semantic=['LM:S:ST2:location:20-40'] | executable=['LM:S:ST2:location:20-40'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST2:location:01:surface-landmark | op=surface\_membership | semantic=['LM:S:ST2:location:7-11'] | executable=['LM:S:ST2:location:7-11'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:ST2:location:20-40 -> RL:S:ST2:location:00:surface-landmark [executable\_to\_relation]

LM:S:ST2:location:7-11 -> RL:S:ST2:location:01:surface-landmark [executable\_to\_relation]

RL:S:ST2:location:01:surface-landmark -> S:ST2:location [relation\_to\_statement]

S:ST2:location -> ST2 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature
- [ ] execution
- [ ] distinct subfeatures
- [ ] distinct executable identity
- [ ] relation operator
- [ ] operand
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# BL40: RESOLVED -> RESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

## WHO source

location | S:BL40:location | page 128 | On the posterior aspect of the knee, at the midpoint of the popliteal crease.

## Semantic decomposition

LM:S:BL40:location:56-76 | source=the popliteal crease | class=crease.skin\_crease | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"1dbdd6b2b621a4ee0185cf93253f7062de8485746c285cadd13ae1177b60ff9","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:popliteal\_crease","subfeature\_type":null} | reason=None

LM:S:BL40:location:7-23 | source=posterior aspect | class=surface.aspect | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"12ffee9bff3884af77a892f4d06484125686f5b1fc7f181d5dc0a994051fc63e","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:posterior\_knee","subfeature\_type":null} | reason=None

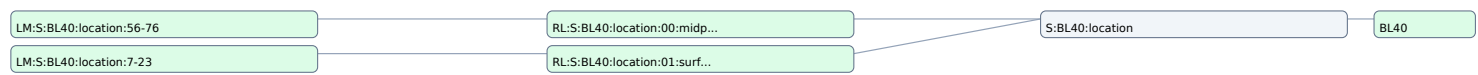
## Composite binding view

## Relation execution

RL:S:BL40:location:00:midpoint-of-entity | op=entity\_midpoint | semantic=['LM:S:BL40:location:56-76'] | executable=['LM:S:BL40:location:56-76'] | dir=None | status=RESOLVED | reason=None

RL:S:BL40:location:01:surface-landmark | op=surface\_membership | semantic=['LM:S:BL40:location:7-23'] | executable=['LM:S:BL40:location:7-23'] | dir=None | status=RESOLVED | reason=None

## Dependency graph



## Dependency graph edge list

LM:S:BL40:location:56-76 -> RL:S:BL40:location:00:midpoint-of-entity [executable\_to\_relation]

RL:S:BL40:location:00:midpoint-of-entity -> S:BL40:location [relation\_to\_statement]

LM:S:BL40:location:7-23 -> RL:S:BL40:location:01:surface-landmark [executable\_to\_relation]

RL:S:BL40:location:01:surface-landmark -> S:BL40:location [relation\_to\_statement]

S:BL40:location -> BL40 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature
- [ ] distinct subfeatures
- [ ] relation operator
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

# ST4: RESOLVED -> UNRESOLVED

Defect family: parent\_only\_composite\_resolution, parent\_only\_composite\_resolution, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, parent\_only\_composite\_resolution, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, parent\_only\_composite\_resolution, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, condition\_linkage\_loss

## WHO source

location | S:ST4:location | page 56 | On the face, 0.4 f-cun lateral to the angle of the mouth.

note | S:ST4:note:1 | page 56 | Note: Lateral to the angle of the mouth, the point is located in the nasolabial sulcus or on the continuation of the nasolabial sulcus.

## Semantic decomposition

LM:S:ST4:location:33-56 | source= the angle of the mouth | class=orifice\_or\_cavity | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST4:location:7-11 | source=face | class=region.body\_region | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA24728","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:ST4:note:1:16-39 | source= the angle of the mouth | class=orifice\_or\_cavity | parent=None | child=None | executor=bound\_composite\_subfeature | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=composite contains subfeature semantics but no unique executable child subfeature geometry

LM:S:ST4:note:1:69-86 | source=nasolabial sulcus | class=soft\_tissue\_feature | parent=None | child=None | executor=approved\_surface\_registry | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=No approved BodyParts3D geometric definition; deliberately unresolved.

LM:S:ST4:note:1:97-134 | source=continuation of the nasolabial sulcus | class=soft\_tissue\_feature | parent=None | child=None | executor=approved\_surface\_registry | status=UNRESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":null,"parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=No approved BodyParts3D geometric definition; deliberately unresolved.

## Composite binding view

## Relation execution

RL:S:ST4:location:00:relative-to | op=directional\_relation | semantic=['LM:S:ST4:location:33-56'] | executable=[] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

RL:S:ST4:location:01:surface-landmark | op=surface\_membership | semantic=['LM:S:ST4:location:7-11'] | executable=['LM:S:ST4:location:7-11'] | dir=None | status=RESOLVED | reason=None

RL:S:ST4:note:1:00:relative-to | op=directional\_relation | semantic=['LM:S:ST4:note:1:16-39'] | executable=[] | dir=None | status=UNRESOLVED | reason=semantic composite has multiple child subfeatures and no unique executable operand

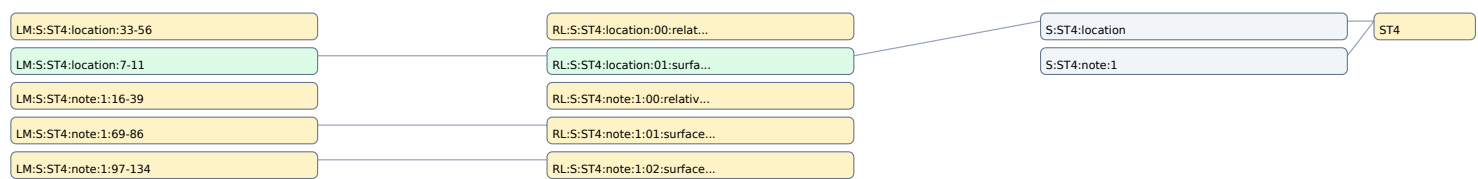
RL:S:ST4:note:1:01:surface-landmark | op=surface\_membership | semantic=['LM:S:ST4:note:1:69-86'] | executable=['LM:S:ST4:note:1:69-86'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

RL:S:ST4:note:1:02:surface-landmark | op=surface\_membership | semantic=['LM:S:ST4:note:1:97-134'] | executable=['LM:S:ST4:note:1:97-134'] | dir=None | status=UNRESOLVED | reason=executable operand not uniquely executable

## Conditions

{"condition\_id":"CD:S:ST4:note:1:00","condition\_type":"alternative","default\_pose\_compatible":null,"reason":"condition preserved as non-executed conditional branch","repaired\_status":"CONDITIONAL","source\_raw":"or on the continuation","source\_statement\_id":"S:ST4:note:1"}

## Dependency graph



## Dependency graph edge list

LM:S:ST4:location:7-11 -> RL:S:ST4:location:01:surface-landmark [executable\_to\_relation]

RL:S:ST4:location:01:surface-landmark -> S:ST4:location [relation\_to\_statement]

LM:S:ST4:note:1:69-86 -> RL:S:ST4:note:1:01:surface-landmark [executable\_to\_relation]

LM:S:ST4:note:1:97-134 -> RL:S:ST4:note:1:02:surface-landmark [executable\_to\_relation]

S:ST4:location -> ST4 [statement\_to\_point]

S:ST4:note:1 -> ST4 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature execution
- [ ] distinct subfeatures distinct executable identity
- [ ] relation operator operand
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT

## TE6: RESOLVED -> RESOLVED

Defect family: relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, direction\_preservation\_failure, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_nonoperand\_semantics\_loss, relation\_semantic\_hash\_mismatch, operand\_trace\_incomplete

### WHO source

location | S:TE6:location | page 169 | On the posterior aspect of the forearm, midpoint of the interosseous space between the radius and the ulna, 3 B-cun proximal to the dorsal wrist crease.

note | S:TE6:note:1 | page 169 | Note: 1 B-cun proximal to TE5, between the radius and the ulna, at the same level as tE7.

### Semantic decomposition

LM:S:TE6:location:127-151 | source= the dorsal wrist crease | class=crease.skin\_crease | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"0ff534739841ae12b463e9ea5102d7801e5de256871212f7ddeb84a0b0265288","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:dorsal\_wrist\_crease","subfeature\_type":null} | reason=None

LM:S:TE6:location:52-106 | source=the interosseous space between the radius and the ulna | class=space.anatomical\_space | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"4b89dd3e588ceb0c6fc9aae7de7f8c71ea842a9e0305158cad97d76f12f35f33","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:radius\_ulna\_interosseous\_space","subfeature\_type":null} | reason=None

LM:S:TE6:location:7-23 | source=posterior aspect | class=surface.aspect | parent=None | child=None | executor=approved\_surface\_registry | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":"7809ff04bb161d448c5b90dbbb407ddc76c10e74e85ee7fe0c795e866ebad73c","kind":"surface\_registry","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":"SR:posterior\_forearm","subfeature\_type":null} | reason=None

LM:S:TE6:location:87-93 | source=radius | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA23463","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:TE6:location:98-106 | source=the ulna | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA23466","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:TE6:note:1:25-29 | source= TE5 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"TE5","registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:TE6:note:1:43-49 | source=radius | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA23463","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:TE6:note:1:54-62 | source=the ulna | class=bone.bone | parent=None | child=None | executor=fma\_mesh | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":"FMA23466","geometry\_hash":null,"kind":"fma\_concept","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":null,"registry\_id":null,"subfeature\_type":null} | reason=None

LM:S:TE6:note:1:85-88 | source=tE7 | class=acupoint\_reference | parent=None | child=None | executor=reference\_acupoint | status=RESOLVED | identity={"concept\_id":null,"constructed\_id":null,"construction\_rule":null,"fma\_id":null,"geometry\_hash":null,"kind":"reference\_acupoint","parent\_landmark\_id":null,"part\_id":null,"reference\_point\_id":"TE7","registry\_id":null,"subfeature\_type":null} | reason=None

### Composite binding view

#### Relation execution

RL:S:TE6:location:00:between | op=between\_constraint | semantic=['LM:S:TE6:location:87-93', 'LM:S:TE6:location:98-106'] | executable=['LM:S:TE6:location:87-93', 'LM:S:TE6:location:98-106'] | dir=None | status=RESOLVED | reason=None

RL:S:TE6:location:01:relative-to | op=directional\_relation | semantic=['LM:S:TE6:location:127-151'] | executable=['LM:S:TE6:location:127-151'] | dir=proximal | status=RESOLVED | reason=None

RL:S:TE6:location:02:midpoint-of-entity | op=entity\_midpoint | semantic=['LM:S:TE6:location:52-106'] | executable=['LM:S:TE6:location:52-106'] | dir=None | status=RESOLVED | reason=None

RL:S:TE6:location:03:surface-landmark | op=surface\_membership | semantic=['LM:S:TE6:location:7-23'] | executable=['LM:S:TE6:location:7-23'] | dir=None | status=RESOLVED | reason=None

RL:S:TE6:note:1:00:between | op=between\_constraint | semantic=['LM:S:TE6:note:1:43-49', 'LM:S:TE6:note:1:54-62'] | executable=['LM:S:TE6:note:1:43-49', 'LM:S:TE6:note:1:54-62'] | dir=None | status=RESOLVED | reason=None

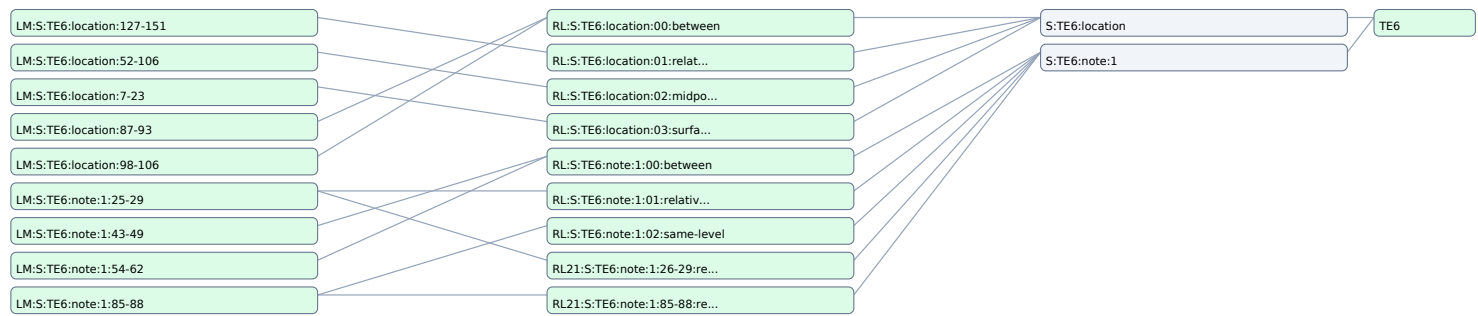
RL:S:TE6:note:1:01:relative-to | op=directional\_relation | semantic=['LM:S:TE6:note:1:25-29'] | executable=['LM:S:TE6:note:1:25-29'] | dir=proximal | status=RESOLVED | reason=None

RL:S:TE6:note:1:02:same-level | op=same\_level\_plane | semantic=['LM:S:TE6:note:1:85-88'] | executable=['LM:S:TE6:note:1:85-88'] | dir=None | status=RESOLVED | reason=None

RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 | op=reference\_dependency | semantic=['LM:S:TE6:note:1:25-29'] | executable=['LM:S:TE6:note:1:25-29'] | dir=None | status=RESOLVED | reason=None

RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e | op=reference\_dependency | semantic=['LM:S:TE6:note:1:85-88'] | executable=['LM:S:TE6:note:1:85-88'] | dir=None | status=RESOLVED | reason=None

Dependency graph



Dependency graph edge list

- LM:S:TE6:location:87-93 -> RL:S:TE6:location:00:between [executable\_to\_relation]
- LM:S:TE6:location:98-106 -> RL:S:TE6:location:00:between [executable\_to\_relation]
- RL:S:TE6:location:00:between -> S:TE6:location [relation\_to\_statement]
- LM:S:TE6:location:127-151 -> RL:S:TE6:location:01:relative-to [executable\_to\_relation]
- RL:S:TE6:location:01:relative-to -> S:TE6:location [relation\_to\_statement]
- LM:S:TE6:location:52-106 -> RL:S:TE6:location:02:midpoint-of-entity [executable\_to\_relation]
- RL:S:TE6:location:02:midpoint-of-entity -> S:TE6:location [relation\_to\_statement]
- LM:S:TE6:location:7-23 -> RL:S:TE6:location:03:surface-landmark [executable\_to\_relation]
- RL:S:TE6:location:03:surface-landmark -> S:TE6:location [relation\_to\_statement]
- LM:S:TE6:note:1:43-49 -> RL:S:TE6:note:1:00:between [executable\_to\_relation]
- LM:S:TE6:note:1:54-62 -> RL:S:TE6:note:1:00:between [executable\_to\_relation]
- RL:S:TE6:note:1:00:between -> S:TE6:note:1 [relation\_to\_statement]
- LM:S:TE6:note:1:25-29 -> RL:S:TE6:note:1:01:relative-to [executable\_to\_relation]
- RL:S:TE6:note:1:01:relative-to -> S:TE6:note:1 [relation\_to\_statement]
- LM:S:TE6:note:1:85-88 -> RL:S:TE6:note:1:02:same-level [executable\_to\_relation]
- RL:S:TE6:note:1:02:same-level -> S:TE6:note:1 [relation\_to\_statement]
- LM:S:TE6:note:1:25-29 -> RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 [executable\_to\_relation]
- RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 -> S:TE6:note:1 [relation\_to\_statement]
- LM:S:TE6:note:1:85-88 -> RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e [executable\_to\_relation]
- RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e -> S:TE6:note:1 [relation\_to\_statement]
- S:TE6:location -> TE6 [statement\_to\_point]
- S:TE6:note:1 -> TE6 [statement\_to\_point]

Coordinates generated: 0 | Legacy C coordinate references: 0

- [ ] WHO source phrase
- [ ] semantic granularity
- [ ] child subfeature
- [ ] distinct subfeatures
- [ ] relation operator
- [ ] condition state
- [ ] RESOLVED/UNRESOLVED
- [ ] ACCEPT
- [ ] REJECT